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(54) **SYSTEM AND METHOD OF DETERMINING OPTIMAL 3-DIMENSIONAL POSITION AND ORIENTATION OF IMAGING DEVICE FOR IMAGING PATIENT BONES**

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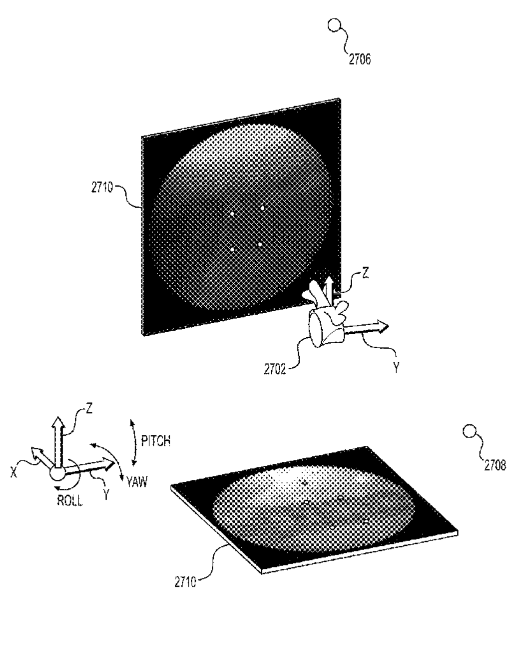
(57) **ABSTRACT**

A method of determining the imaging arm's optimal 3-dimensional position and orientation for taking images of a body implant or body structure such as vertebral body is provided. Test images of vertebral body of interest are initially taken by the user and are received by the imaging device. The test images typically include AP and lateral x-ray images of the vertebral body. From the test images, the vertebral body is segmented. A 3-dimensional model of the vertebral body is then aligned against the corresponding vertebral body in the test images. Based on the alignment, a 3-dimensional position and orientation of the imaging arm for taking optimal A-P and lateral x-ray images are determined based on the aligned 3-dimensional model. The present method eliminates the need to repeatedly take fluoro shots manually to find the optimum images to thereby reduce procedural time, x-ray exposure and procedure costs.

22 Claims, 31 Drawing Sheets



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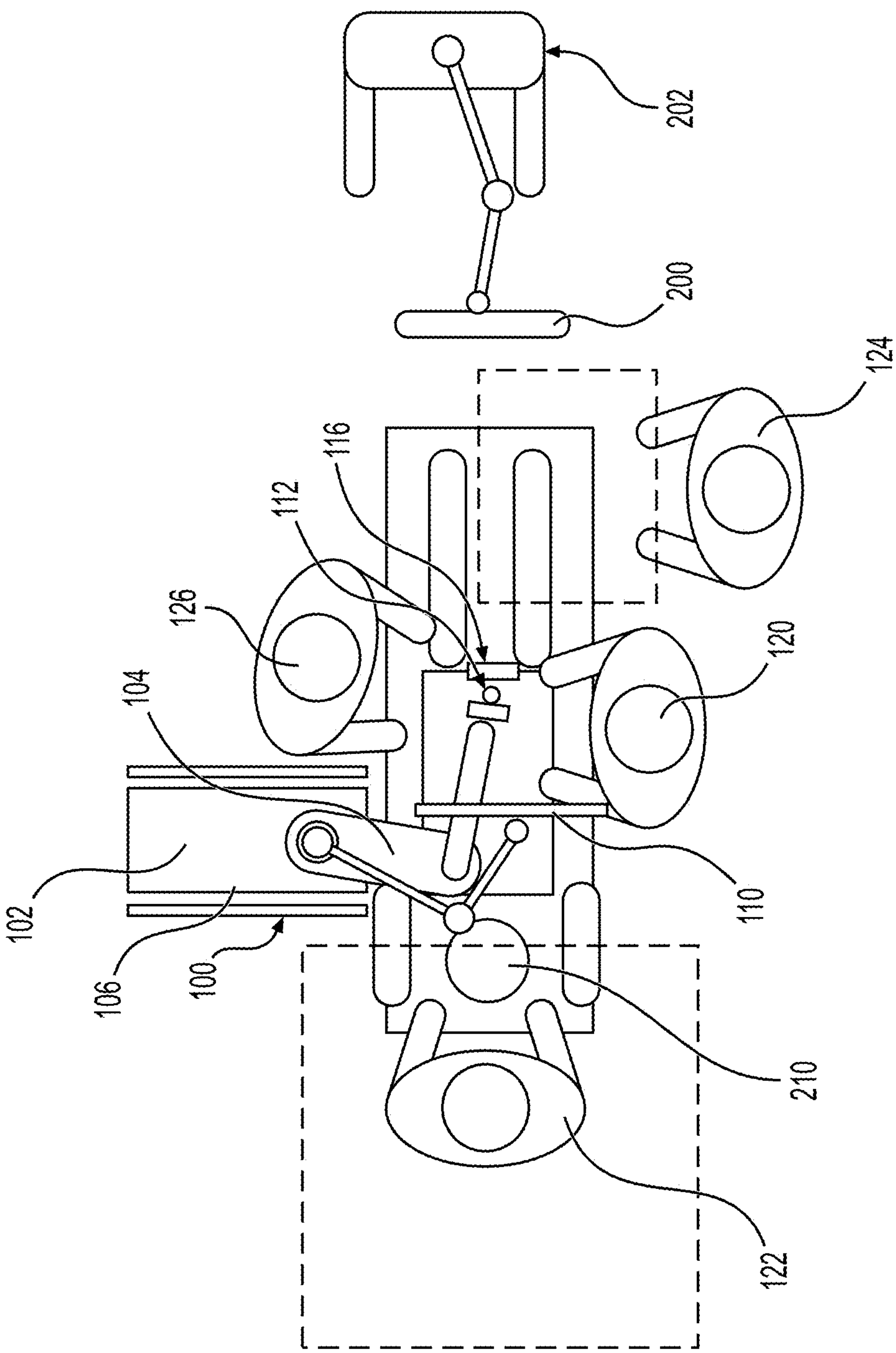


FIG. 1

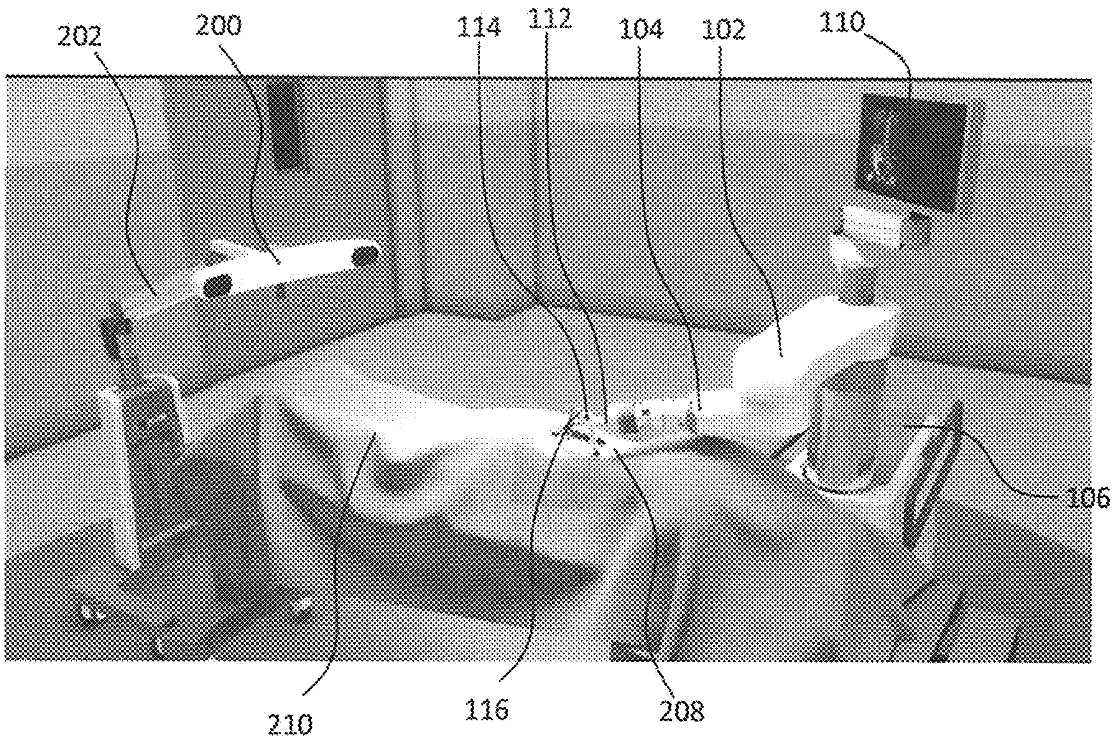
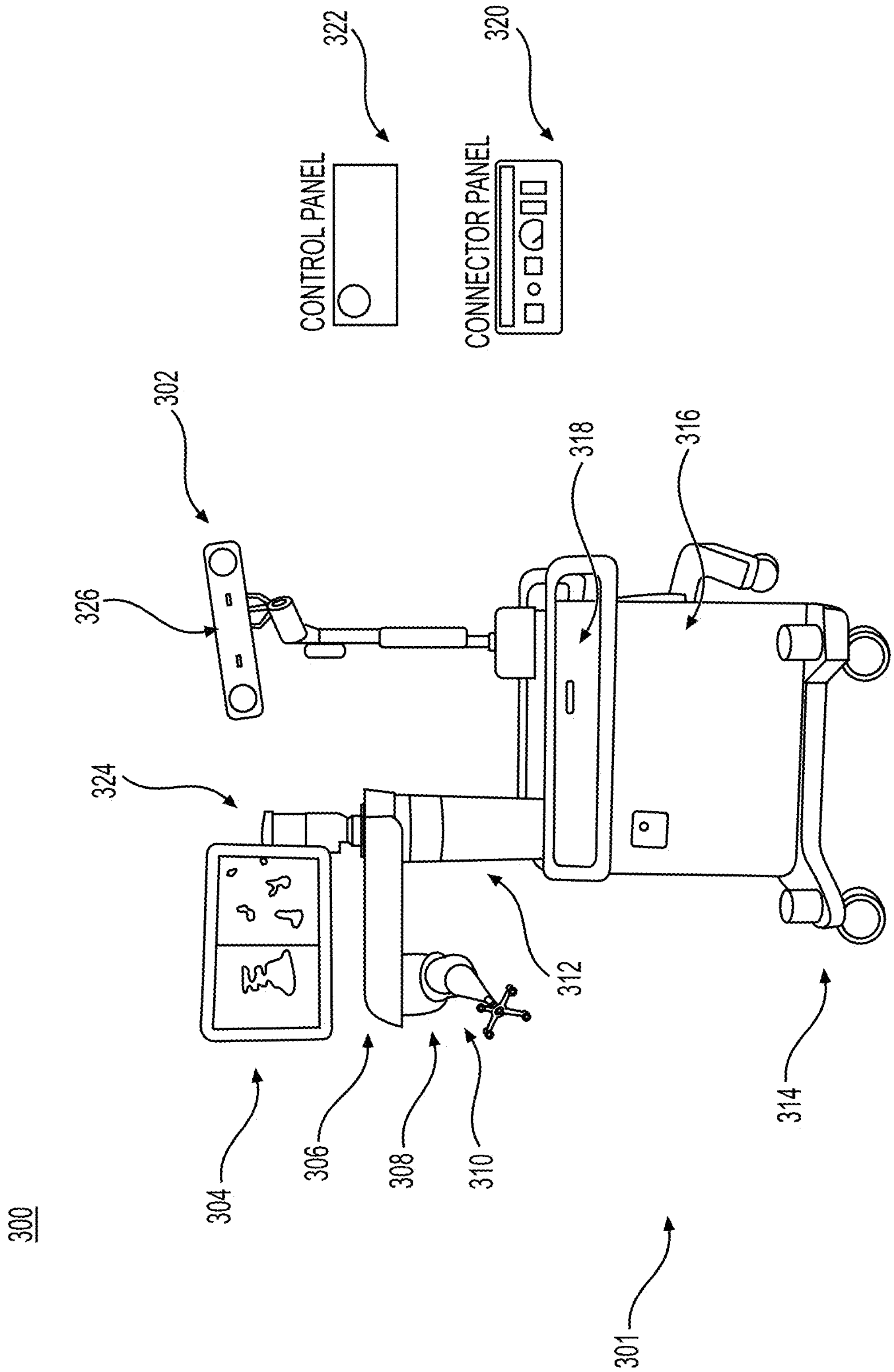


FIG. 2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326
CONTROL PANEL
CONNECTOR PANEL

FIG. 3

400

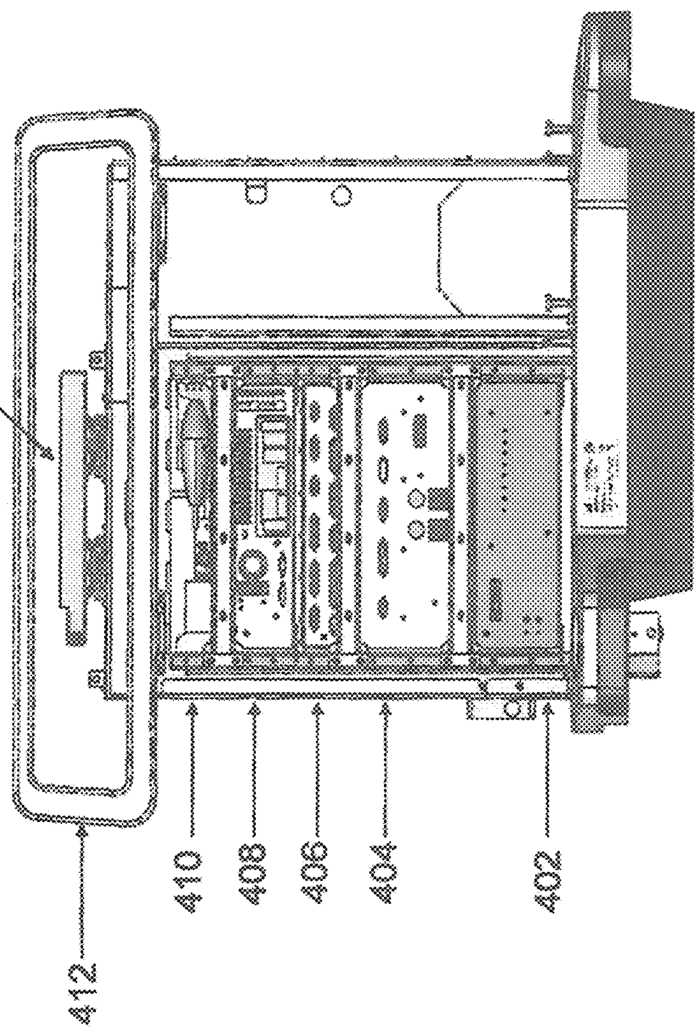
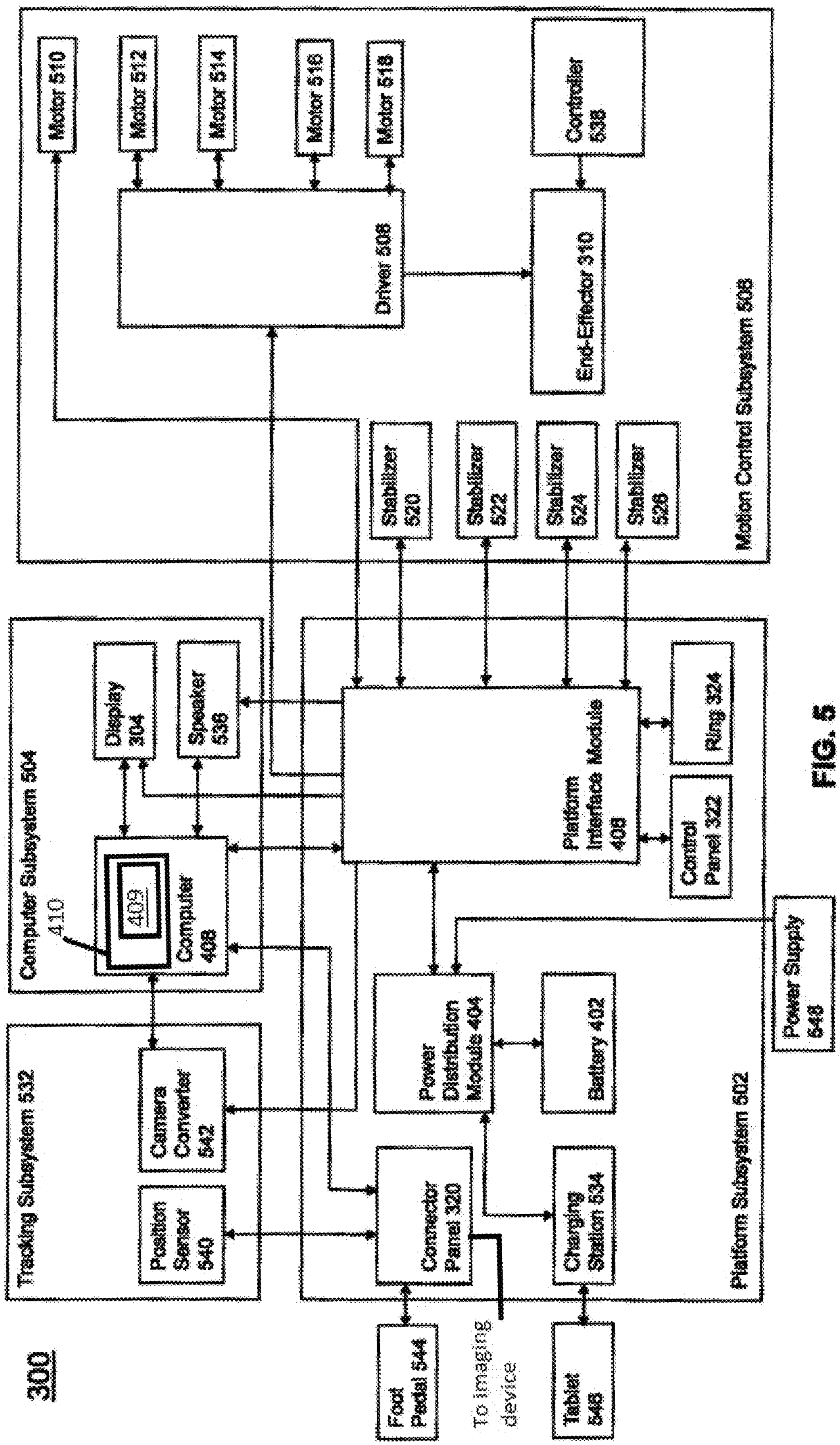


FIG. 4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Motor 510
Motor 512
Motor 514
Motor 516
Motor 518
Controller 538
Driver 508
End-Effector 310
Stabilizer 520
Stabilizer 522
Stabilizer 524
Stabilizer 526
Motion Control Subsystem 506
Computer Subsystem 504
Display 304
Speaker 536
410
409
Computer 408
Platform Interface Module 406
Ring 324
Control Panel 322
Tracking Subsystem 532
Camera Converter 542
Position Sensor 540
Power Distribution Module 404
Battery 402
Power Supply 548
Connector Panel 320
Charging Station 534
Platform Subsystem 502
Foot Pedal 544
To imaging device
Tablet 546

FIG. 5

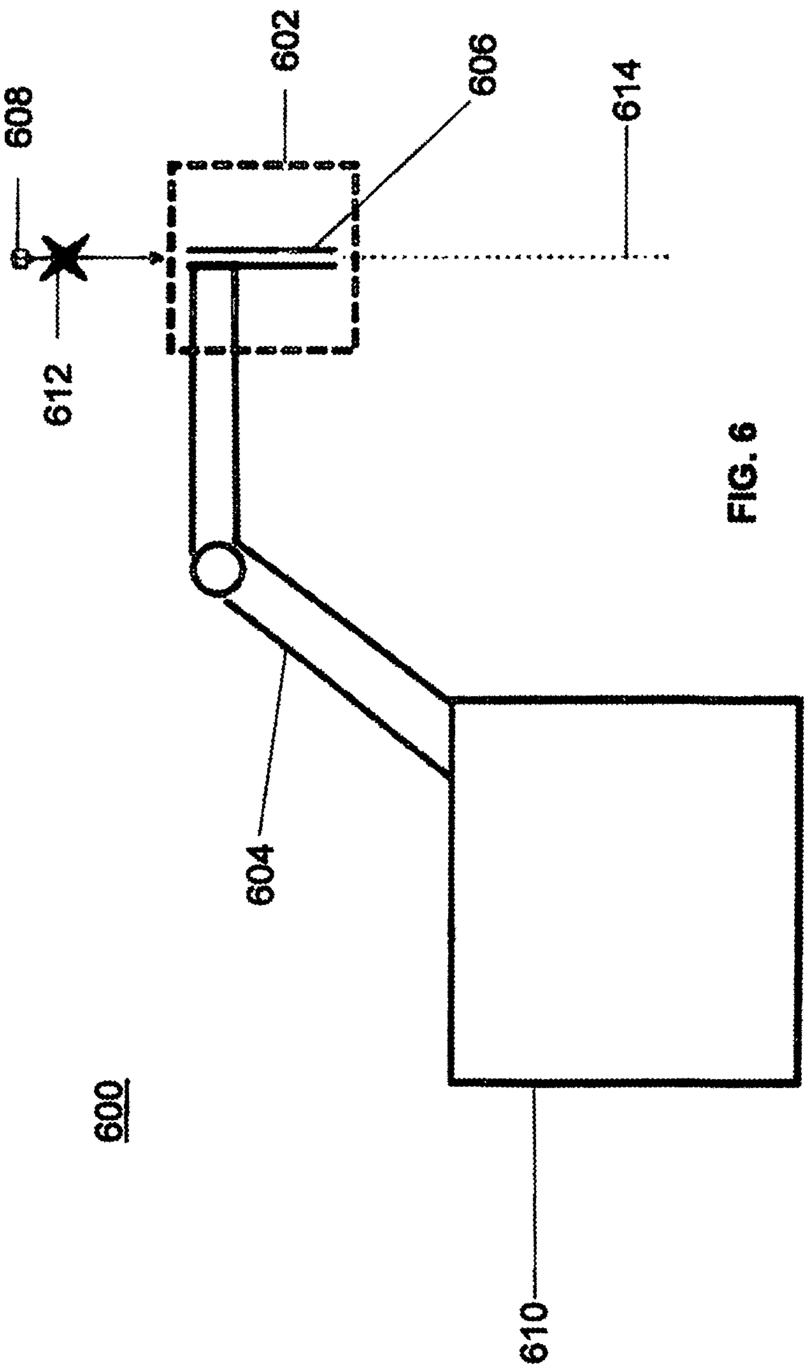

FIG. 6

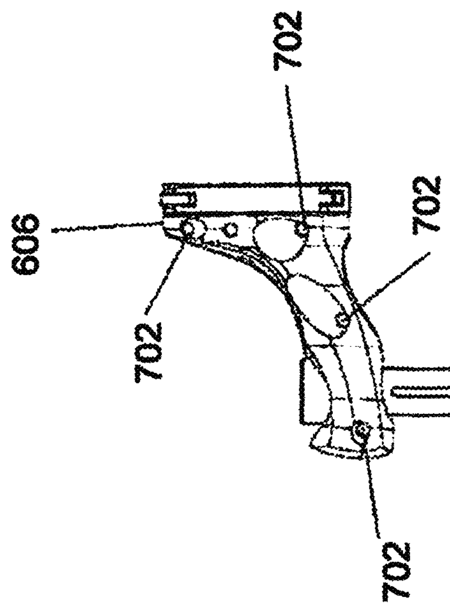

FIG. 7C

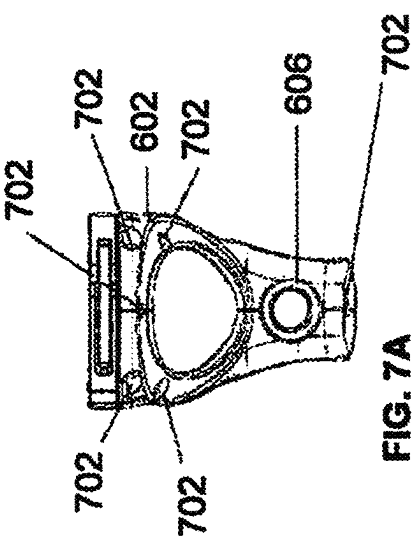

FIG. 7A

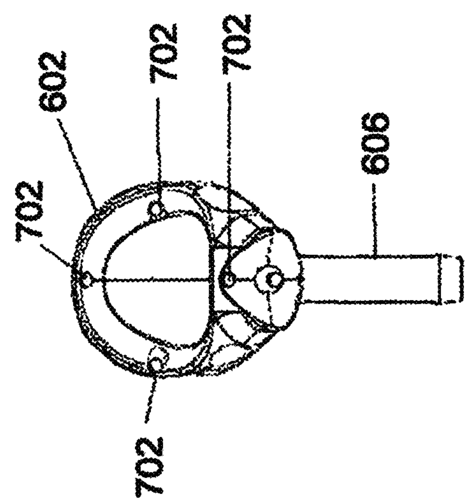

FIG. 7B

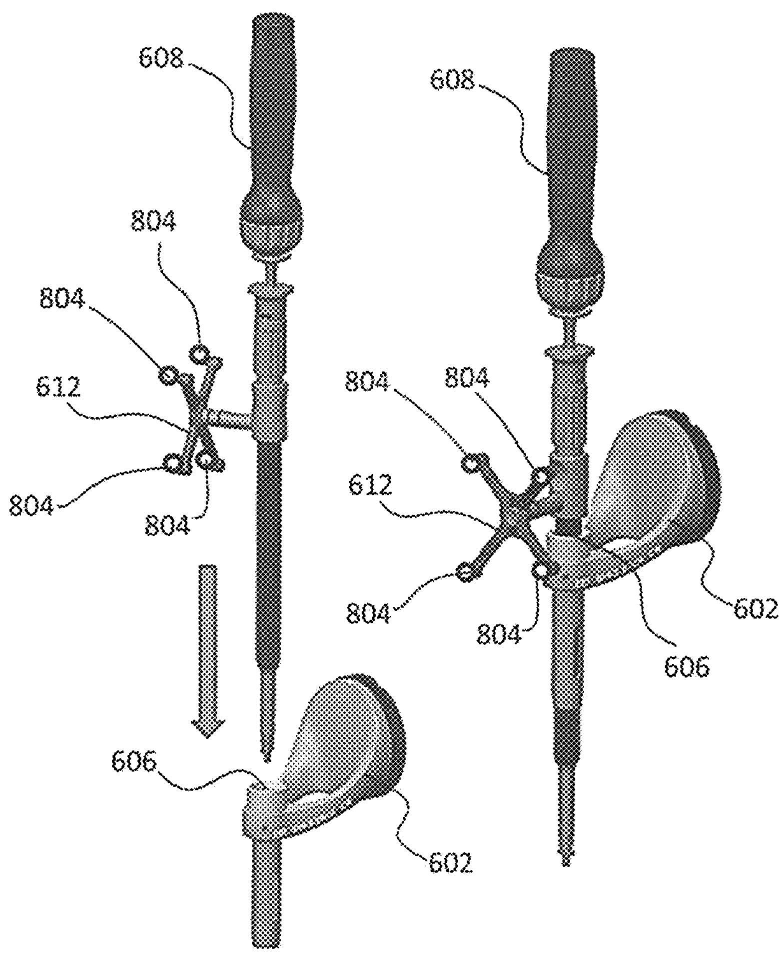

FIG. 8

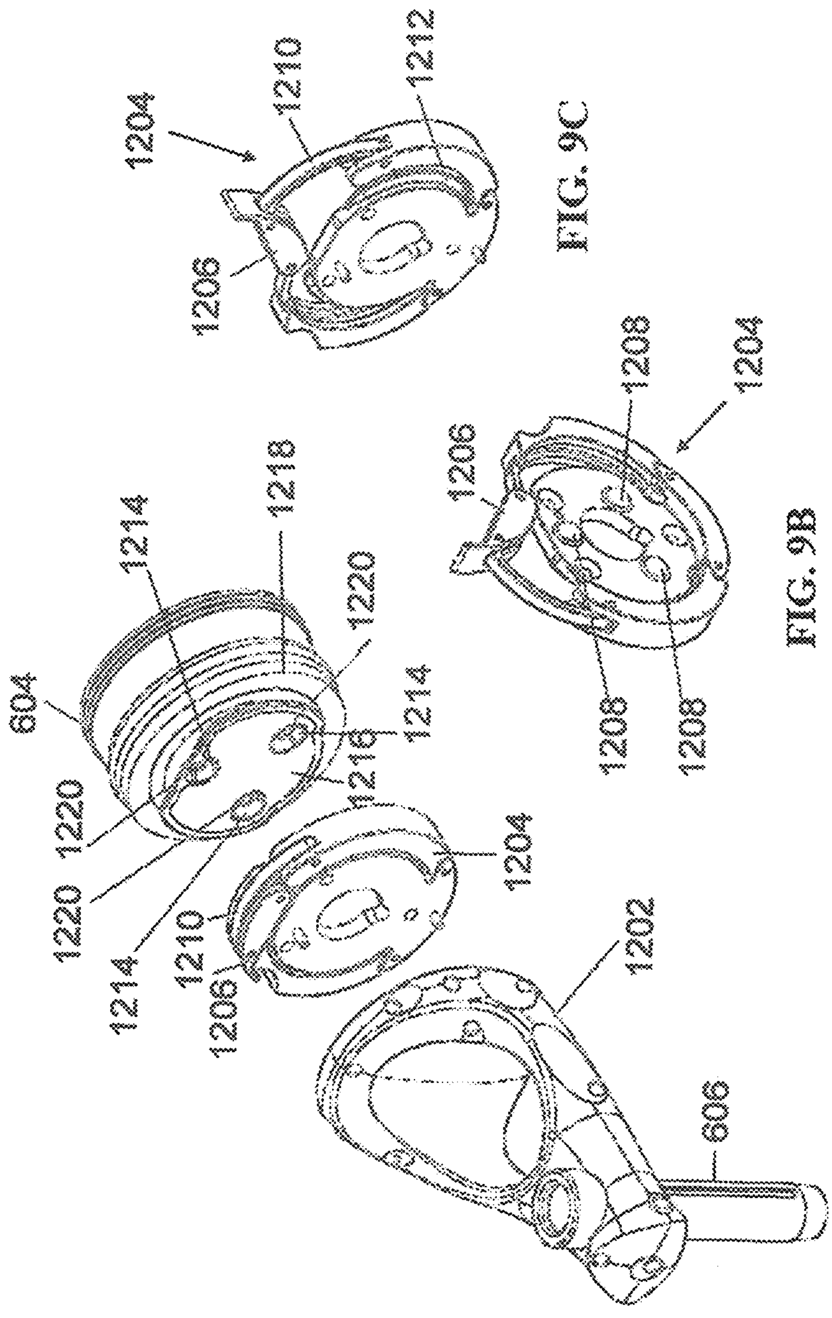

FIG. 9C

FIG. 9B

FIG. 9A

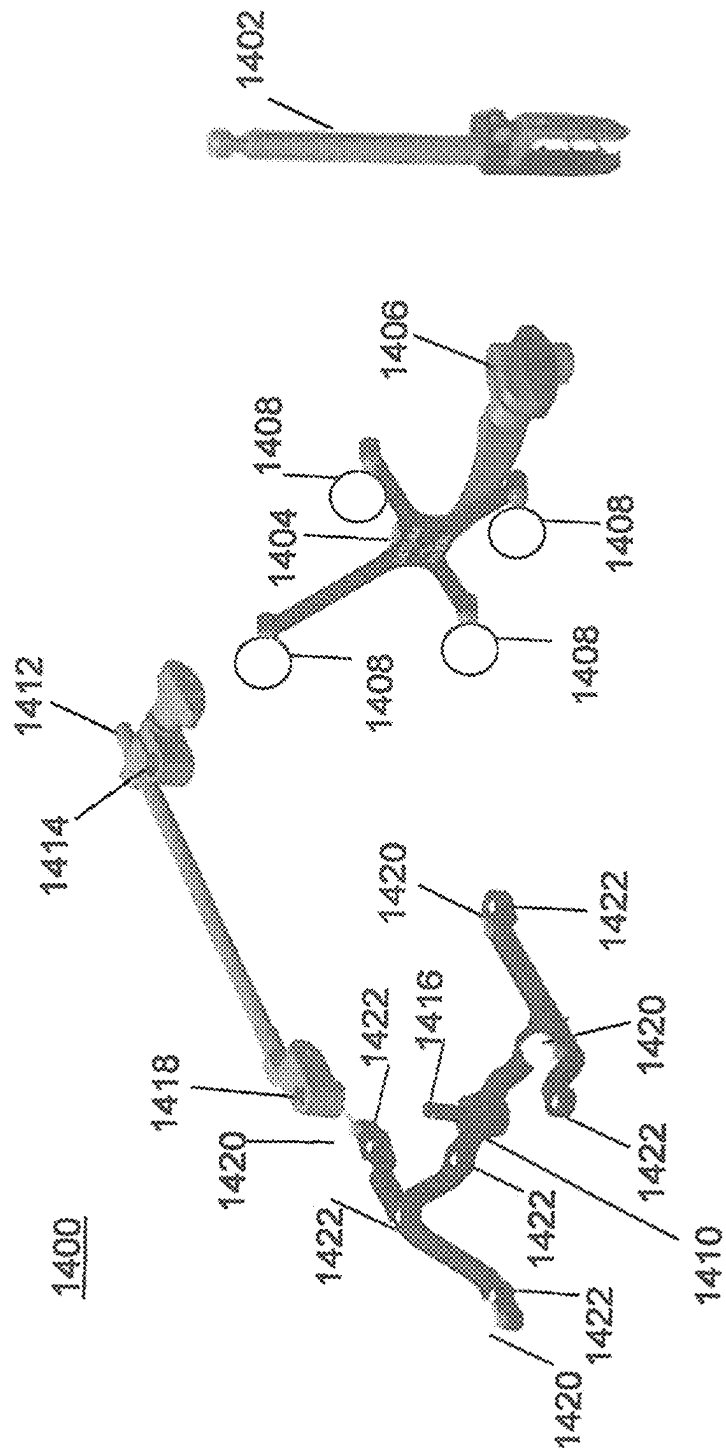
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FIG. 10

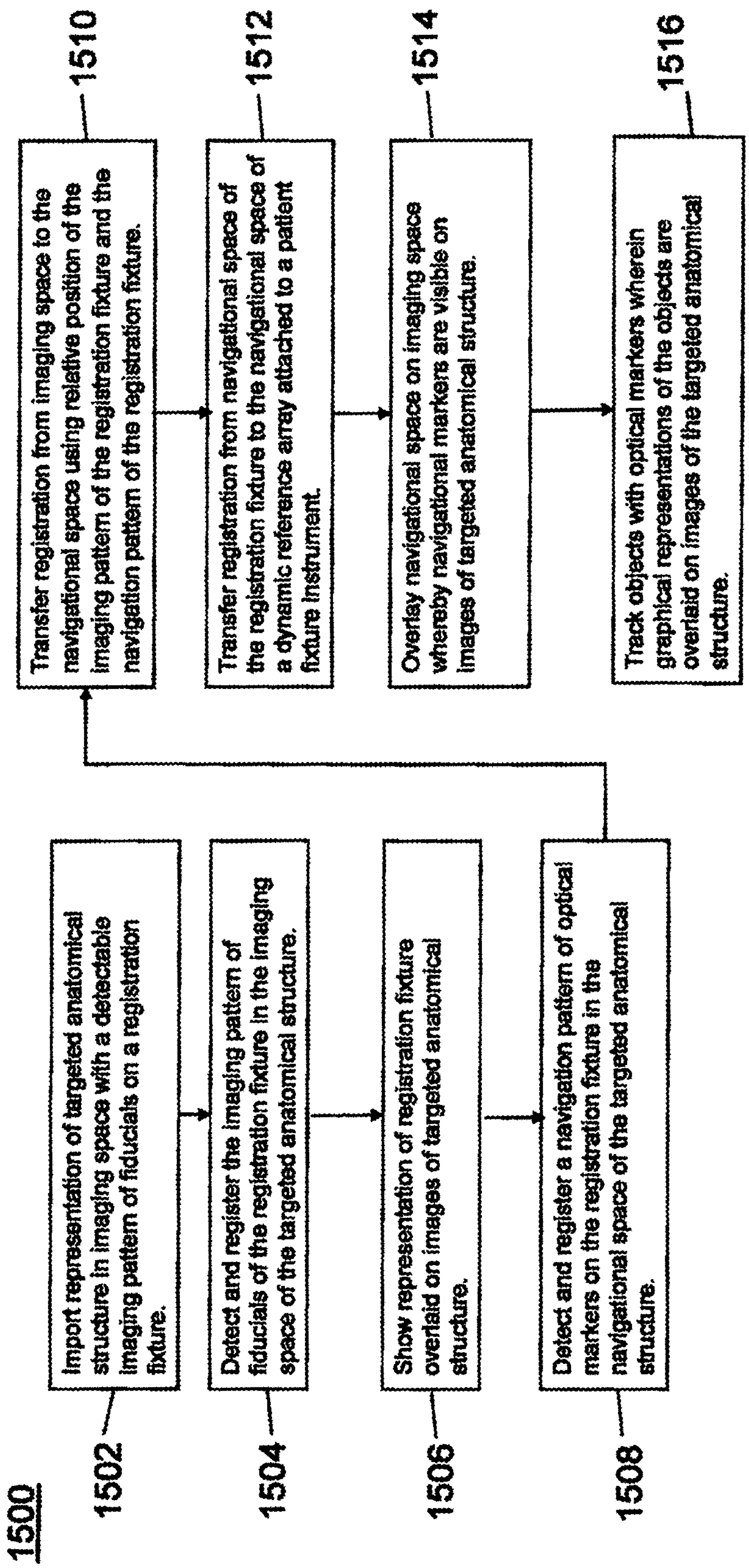

1500
1502
Import representation of targeted anatomical structure in imaging space with a detectable imaging pattern of fiducials on a registration fixture.
1504
Detect and register the imaging pattern of fiducials of the registration fixture in the imaging space of the targeted anatomical structure.
1506
Show representation of registration fixture overlaid on images of targeted anatomical structure.
1508
Detect and register a navigation pattern of optical markers on the registration fixture in the navigational space of the targeted anatomical structure.
1510
Transfer registration from imaging space to the navigational space using relative position of the imaging pattern of the registration fixture and the navigation pattern of the registration fixture.
1512
Transfer registration from navigational space of the registration fixture to the navigational space of a dynamic reference array attached to a patient fixture instrument.
1514
Overlay navigational space on imaging space whereby navigational markers are visible on images of targeted anatomical structure.
1516
Track objects with optical markers wherein graphical representations of the objects are overlaid on images of the targeted anatomical structure.

FIG. 11

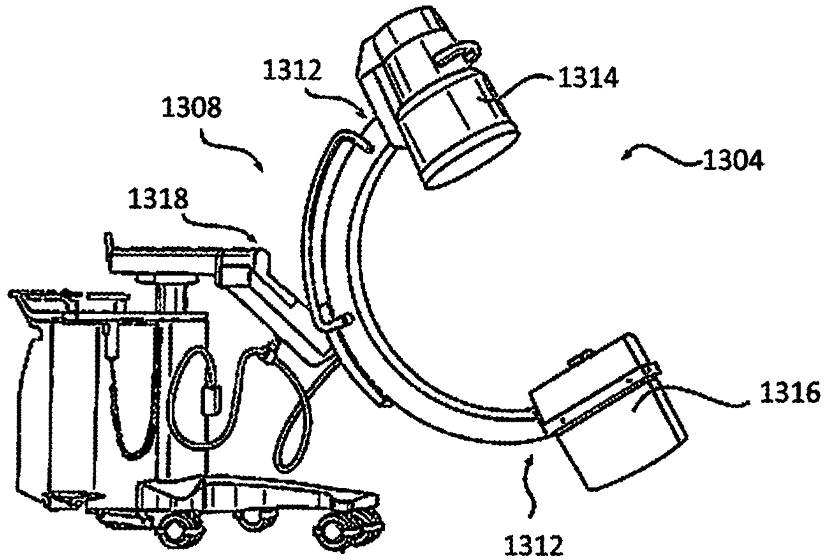


FIG. 12A

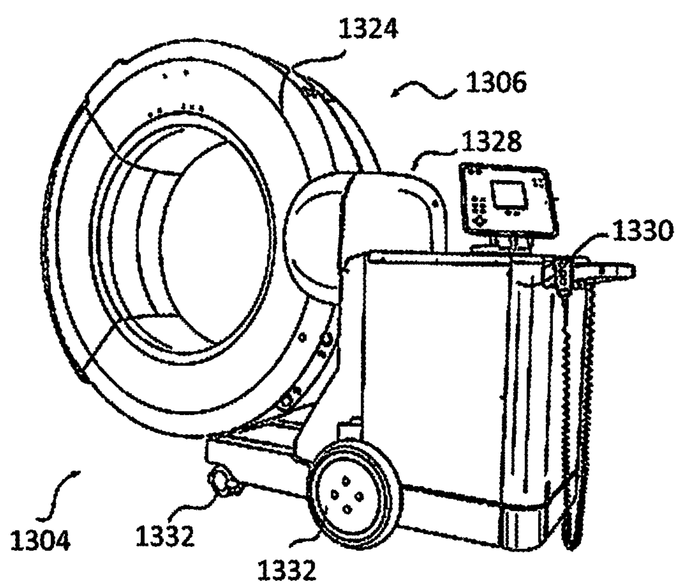


FIG. 12B

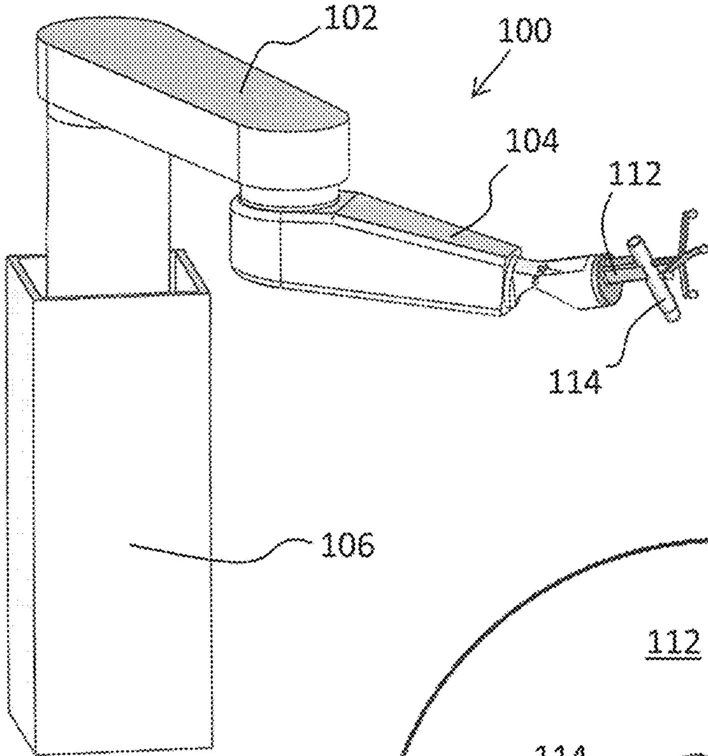


FIG. 13A

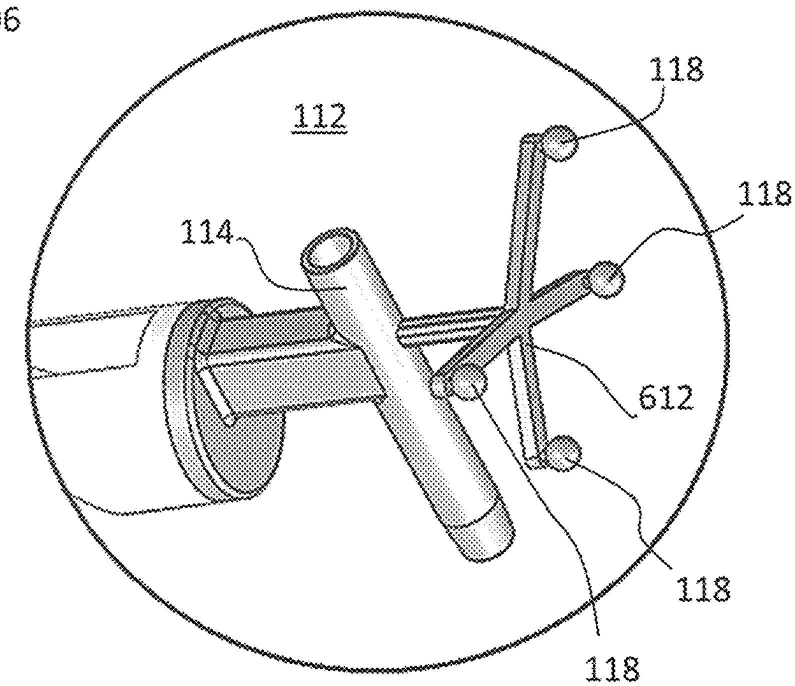


FIG. 13B

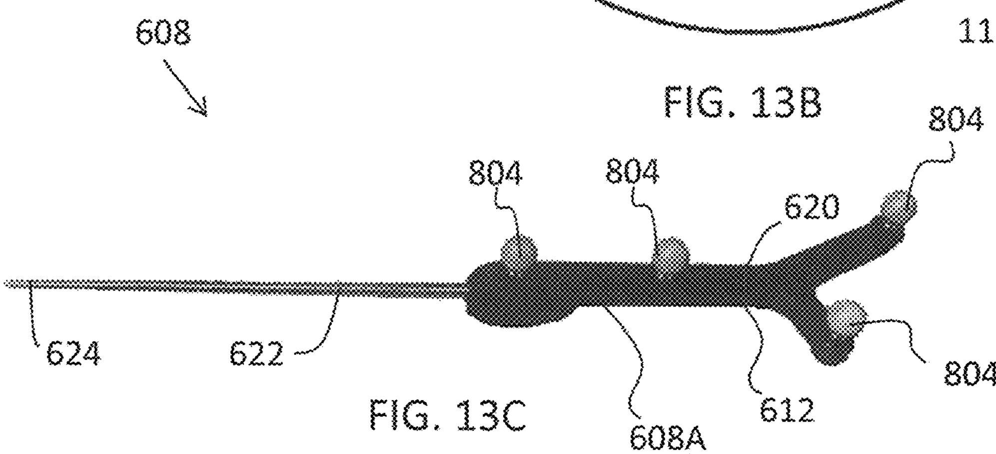


FIG. 13C

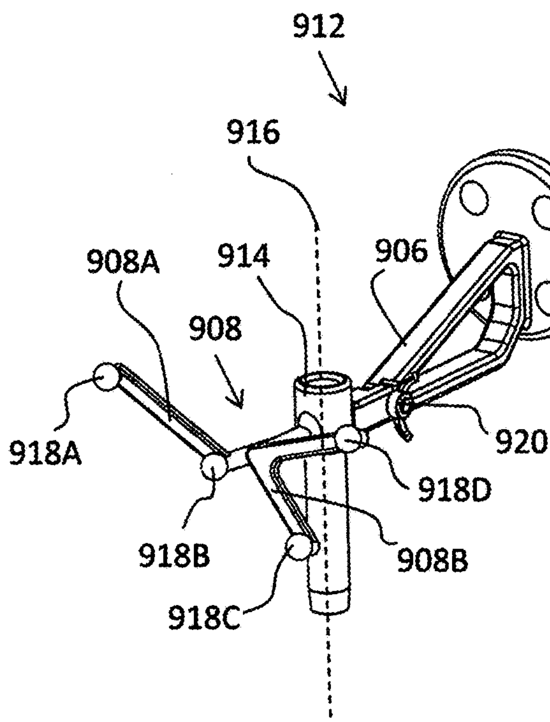


FIG. 14A

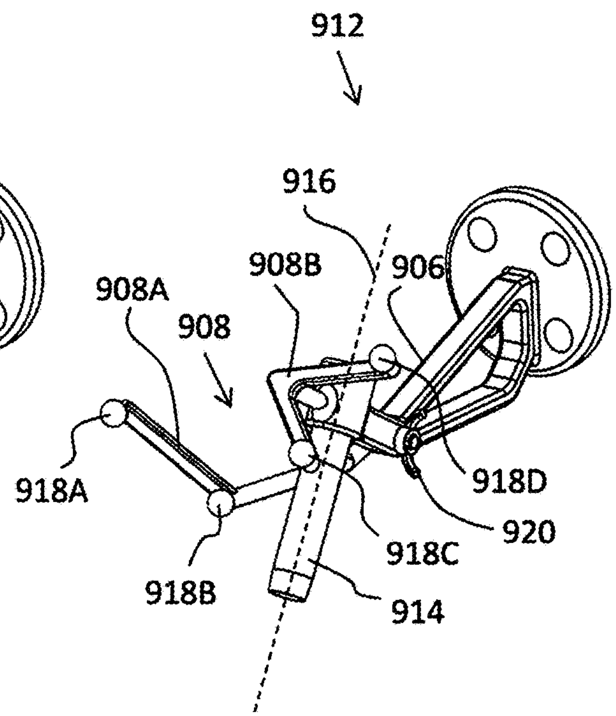


FIG. 14B

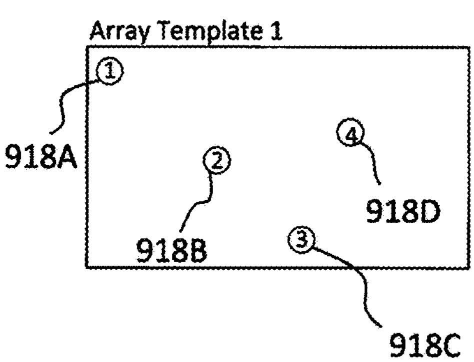


FIG. 14C

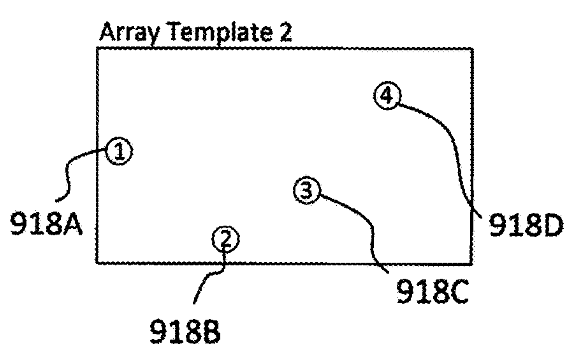


FIG. 14D

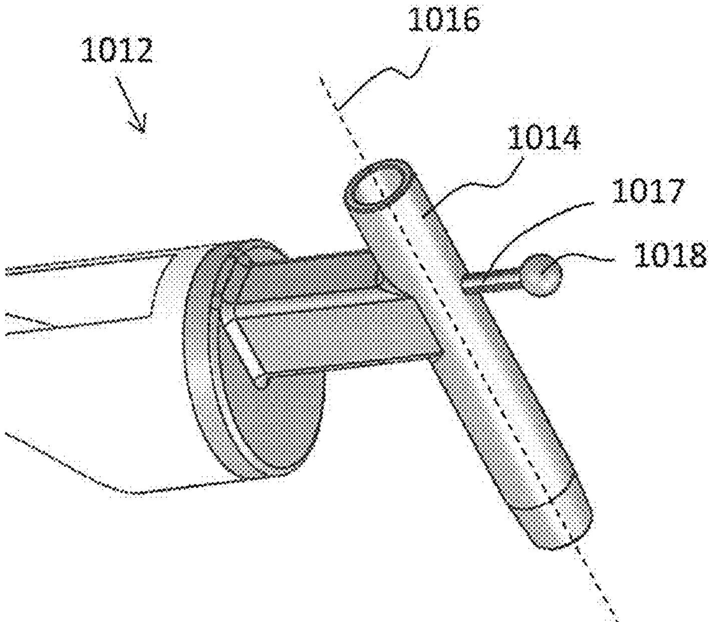


FIG. 15A

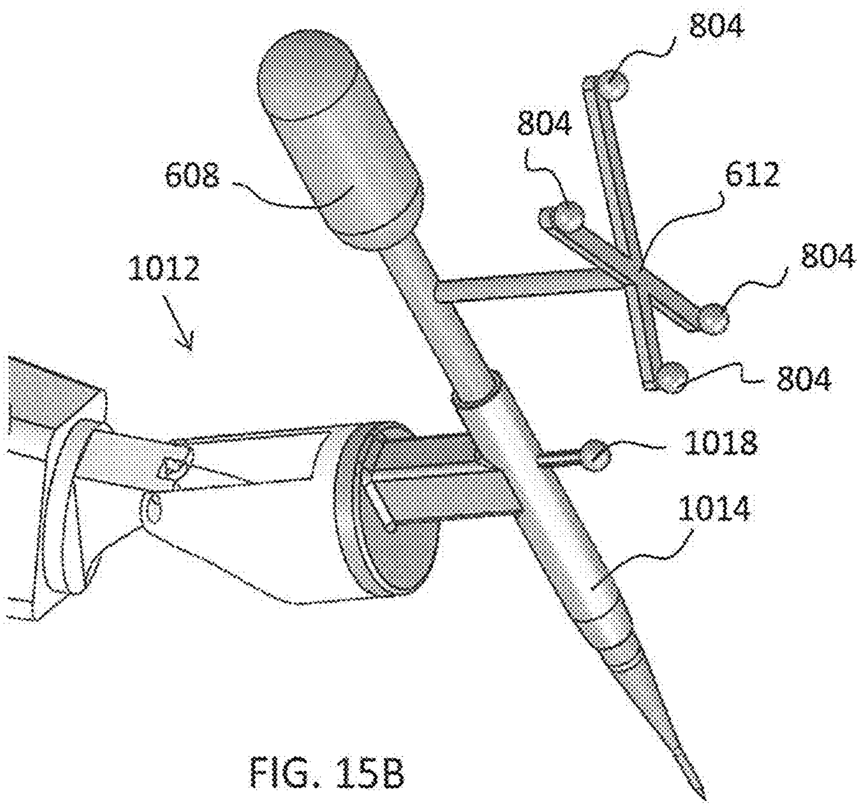
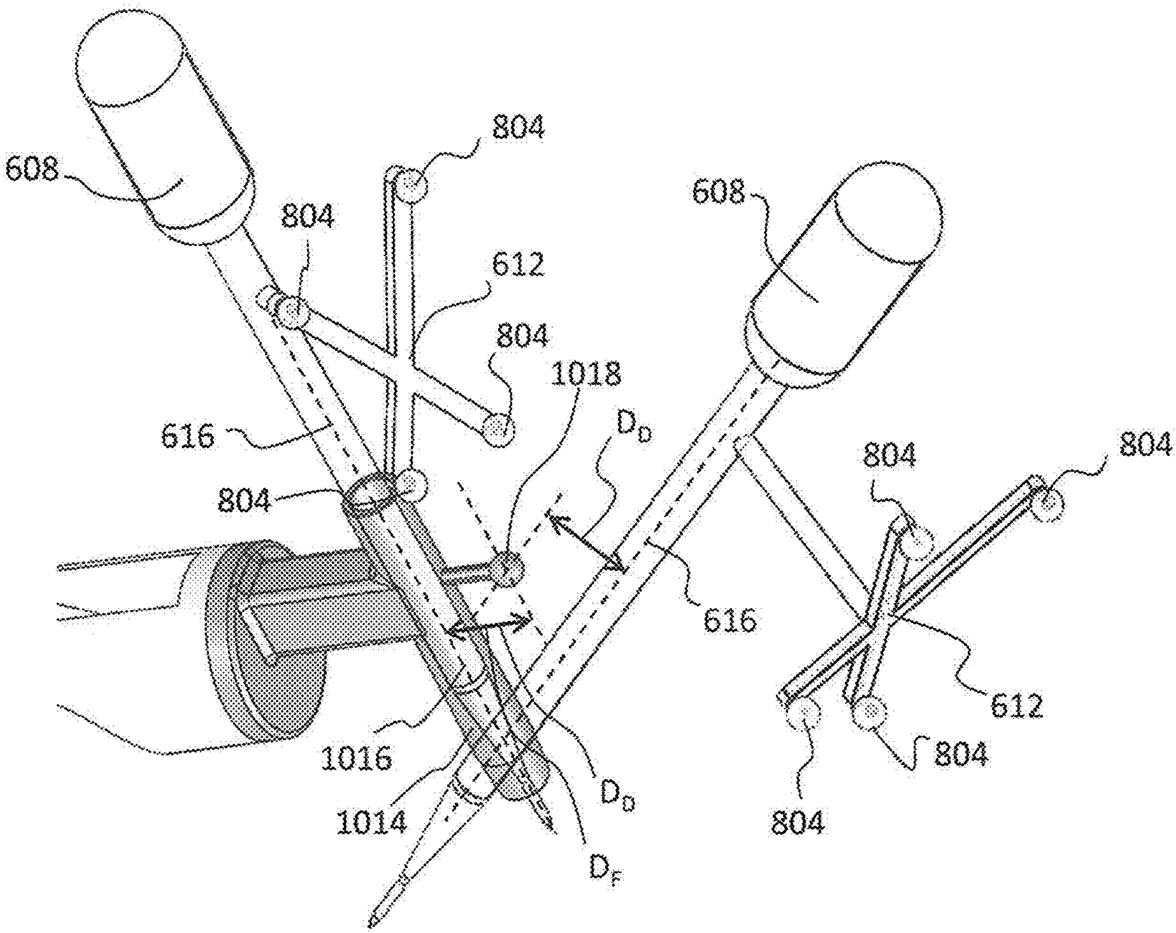


FIG. 15B



608
804
804
612
804
1018
608
616
D_D
804
804
804
616
612
1016
804
1014
D_D
804
D_F

FIG. 15C

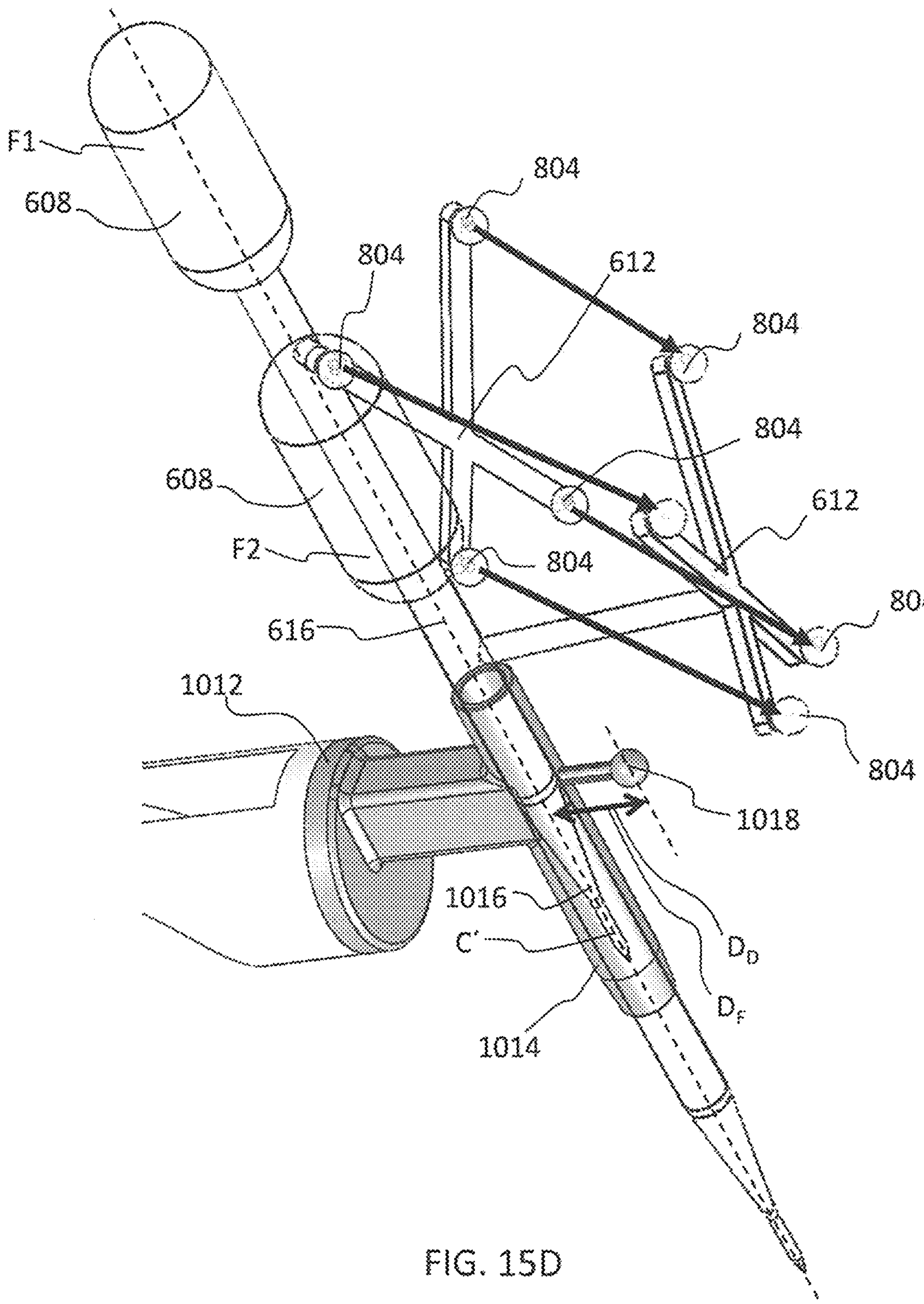

F1
608
804
612
804
804
804
608
612
F2
804
804
616
1012
804
1018
1016
C'
D_D
D_F
1014

FIG. 15D

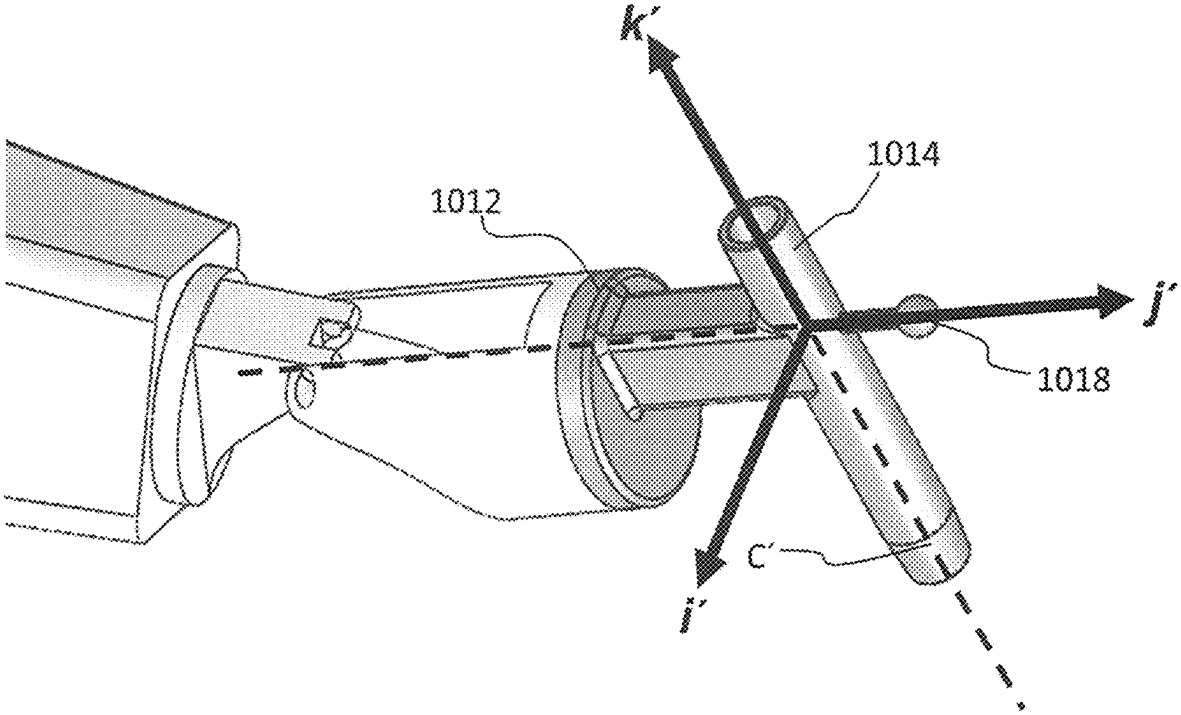


FIG. 15E

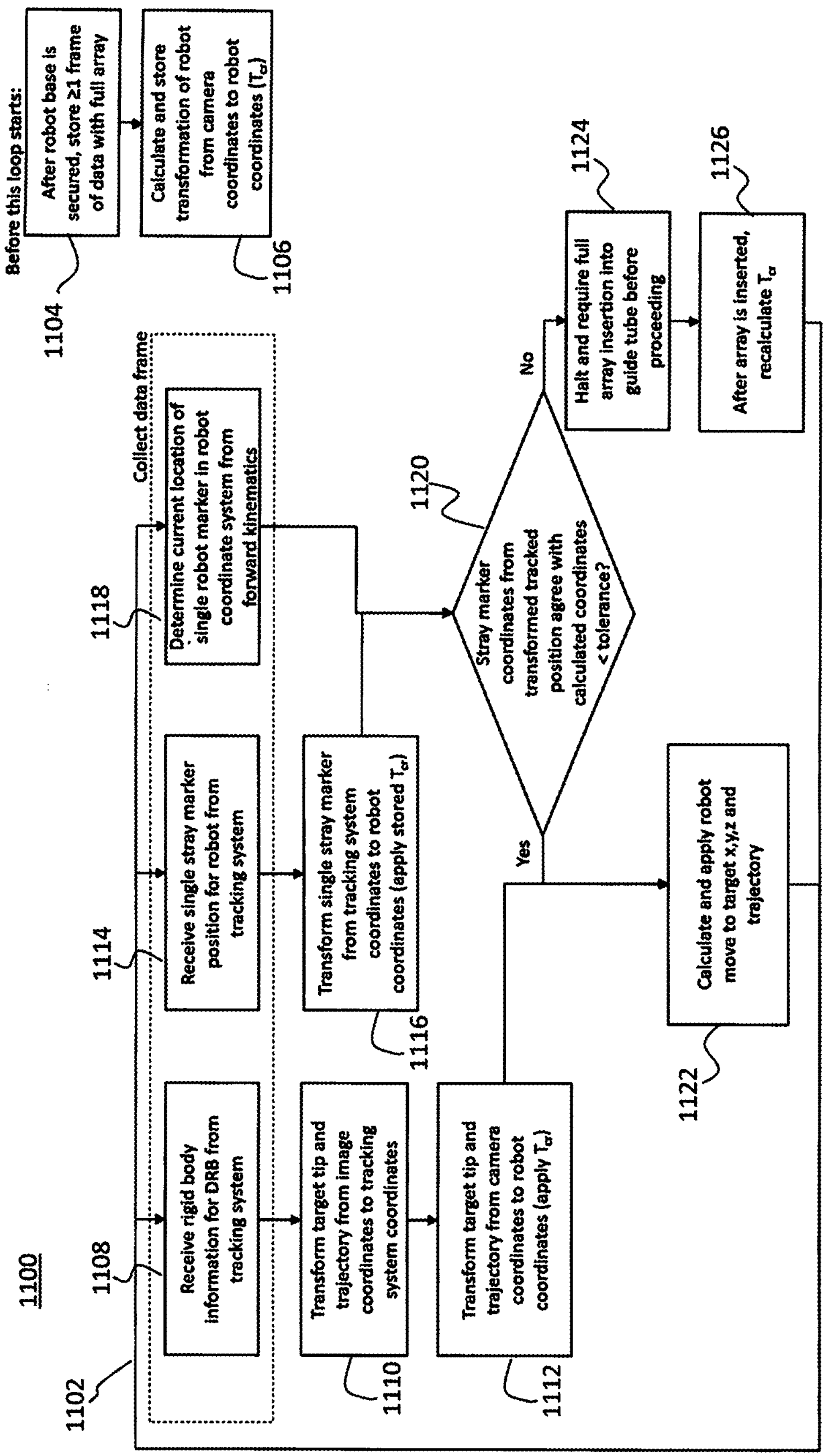

1100
Before this loop starts:
After robot base is secured, store ≥1 frame of data with full array
1104
Calculate and store transformation of robot from camera coordinates to robot coordinates (T_{cr})
1106
1102
Collect data frame
1108
Receive rigid body information for DRB from tracking system
1114
Receive single stray marker position for robot from tracking system
1118
Determine current location of single robot marker in robot coordinate system from forward kinematics
Transform target tip and trajectory from image coordinates to tracking system coordinates
1110
Transform single stray marker from tracking system coordinates to robot coordinates (apply stored T_{cr})
1116
Transform target tip and trajectory from camera coordinates to robot coordinates (apply T_{cr})
1112
1120
Stray marker coordinates from transformed tracked position agree with calculated coordinates < tolerance?
No
Yes
Halt and require full array insertion into guide tube before proceeding
1124
After array is inserted, recalculate T_{cr}
1126
Calculate and apply robot move to target x,y,z and trajectory
1122

FIG. 16

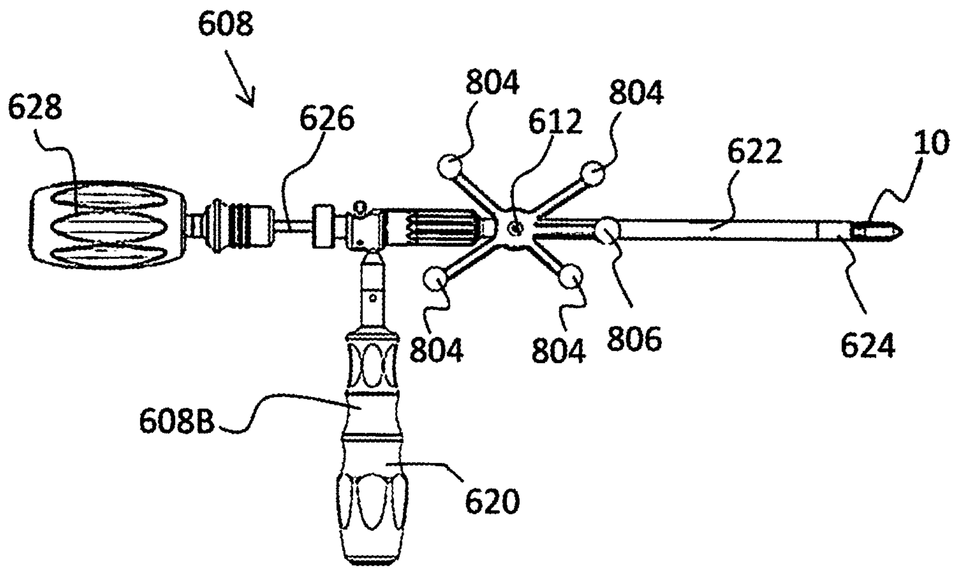


FIG. 17A

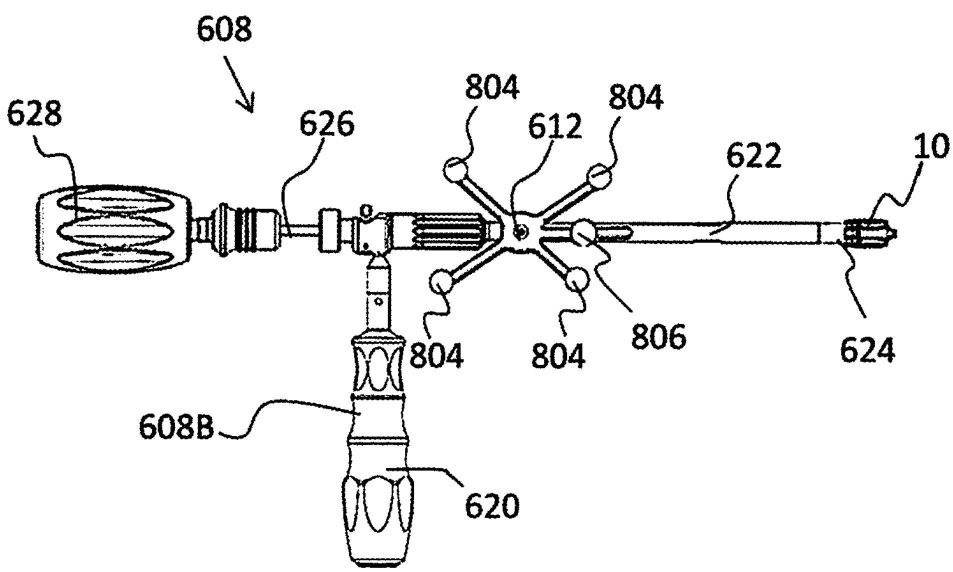


FIG. 17B

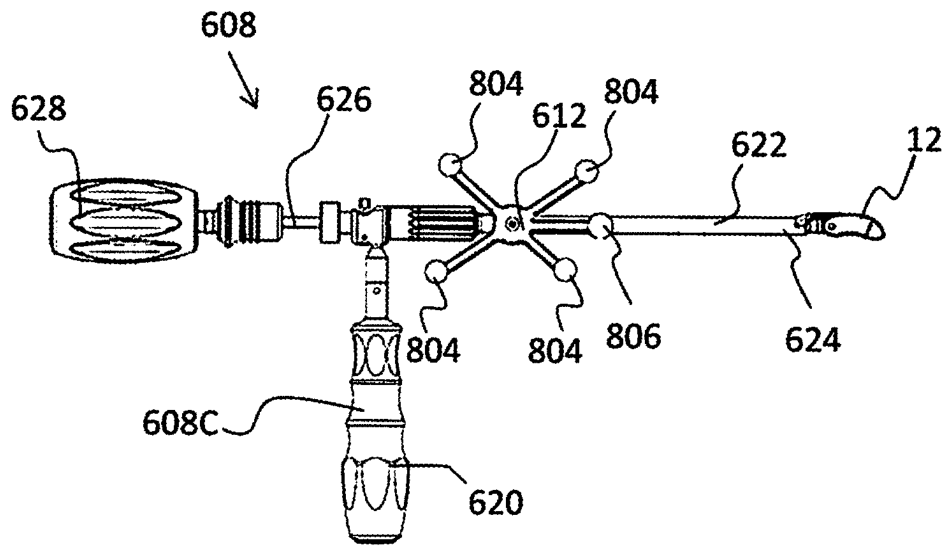


FIG. 18A

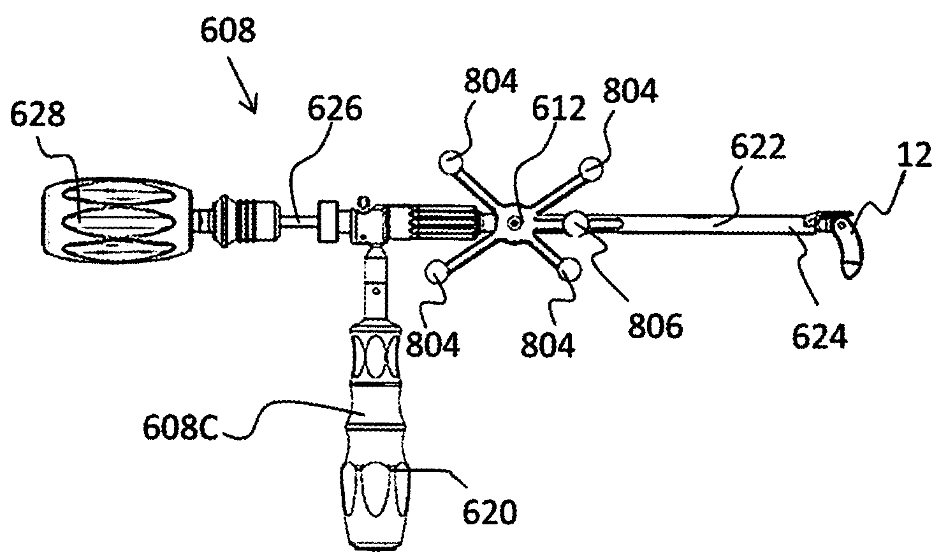


FIG. 18B

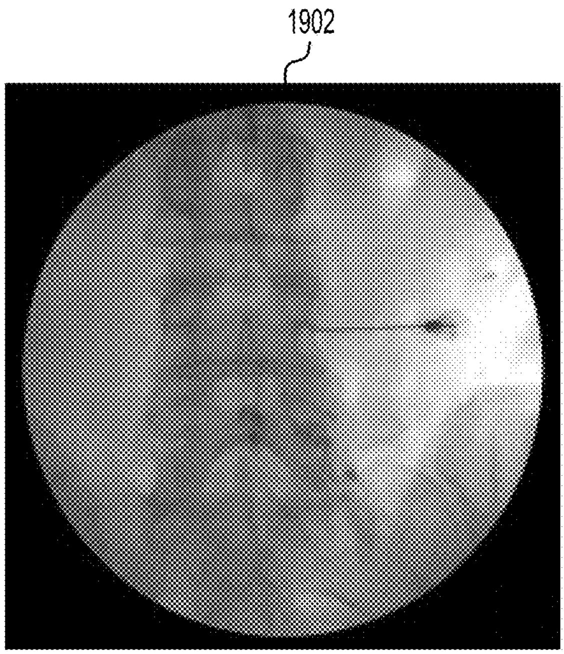


FIG. 19A

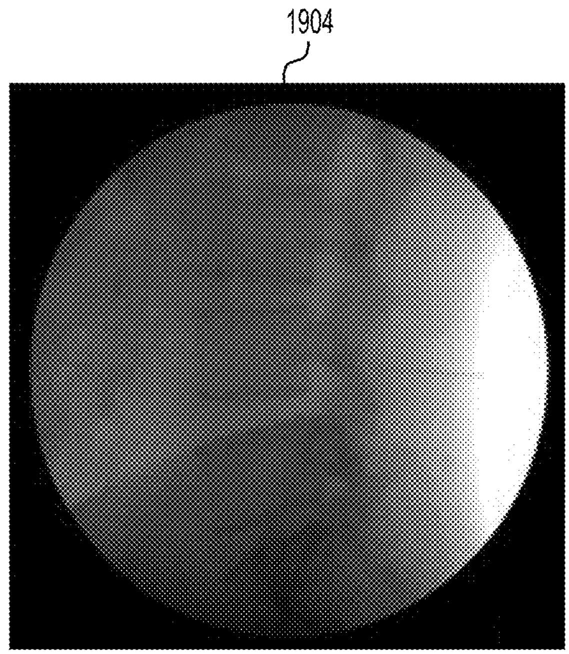


FIG. 19B

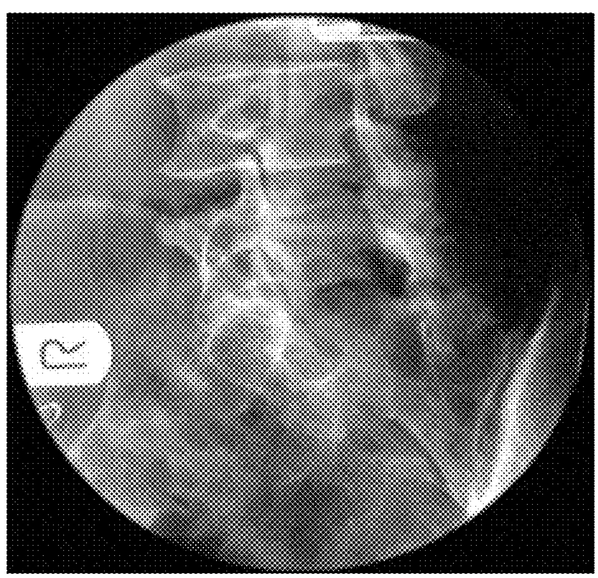


FIG. 20

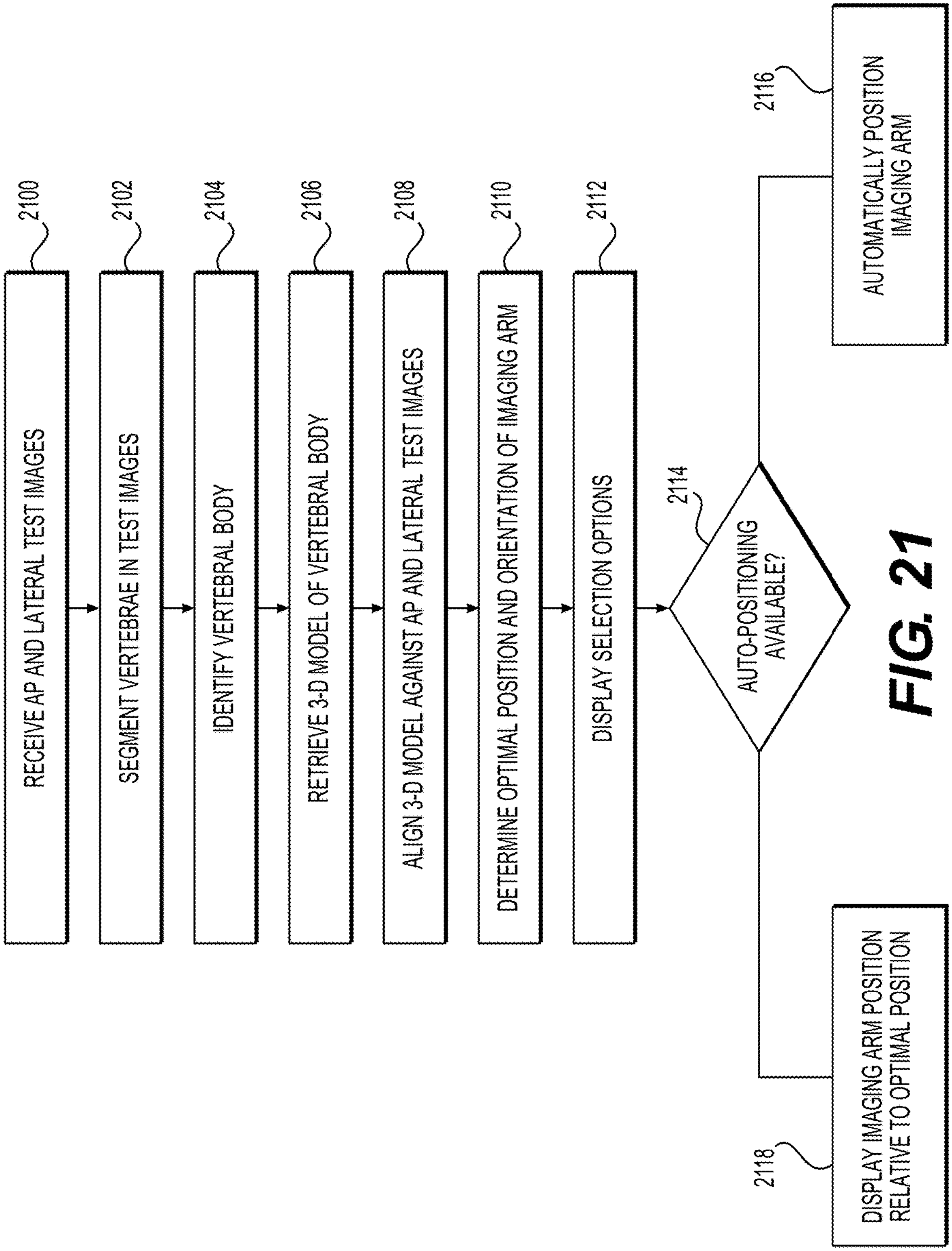

2100
RECEIVE AP AND LATERAL TEST IMAGES
2102
SEGMENT VERTEBRAE IN TEST IMAGES
2104
IDENTIFY VERTEBRAL BODY
2106
RETRIEVE 3-D MODEL OF VERTEBRAL BODY
2108
ALIGN 3-D MODEL AGAINST AP AND LATERAL TEST IMAGES
2110
DETERMINE OPTIMAL POSITION AND ORIENTATION OF IMAGING ARM
2112
DISPLAY SELECTION OPTIONS
2114
AUTO-POSITIONING AVAILABLE?
2116
AUTOMATICALLY POSITION IMAGING ARM
2118
DISPLAY IMAGING ARM POSITION RELATIVE TO OPTIMAL POSITION

FIG. 21

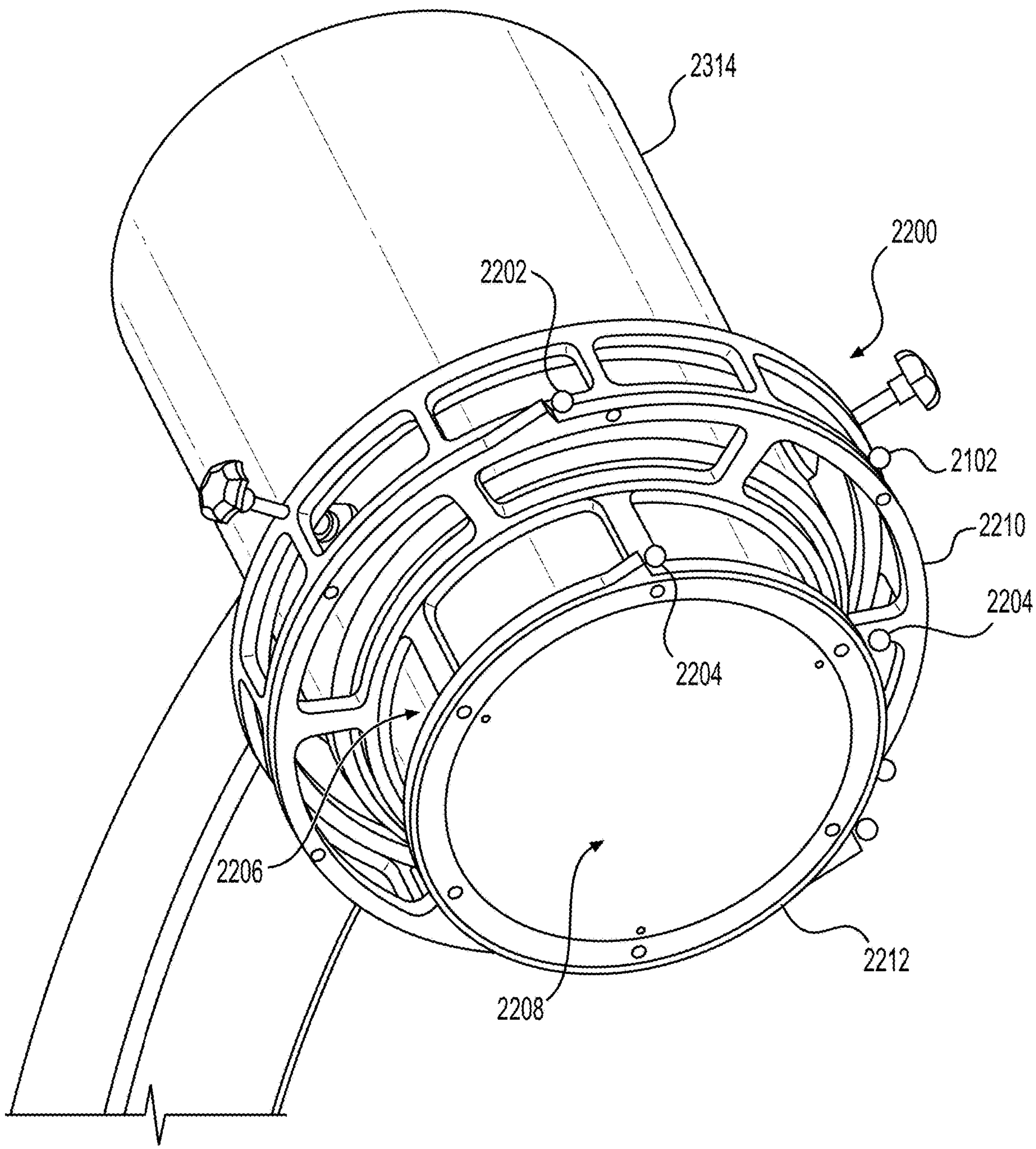


FIG. 22

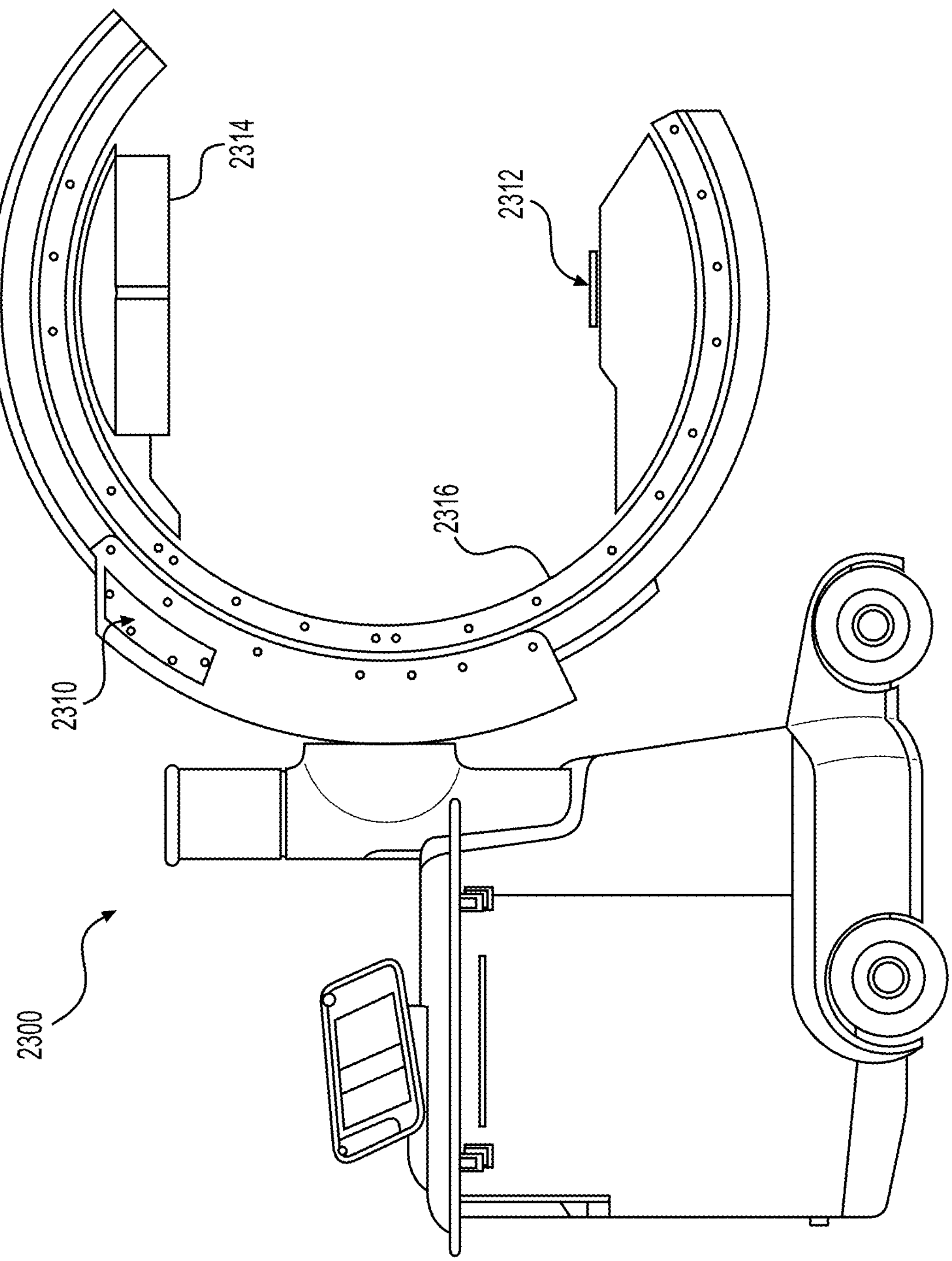


FIG. 23

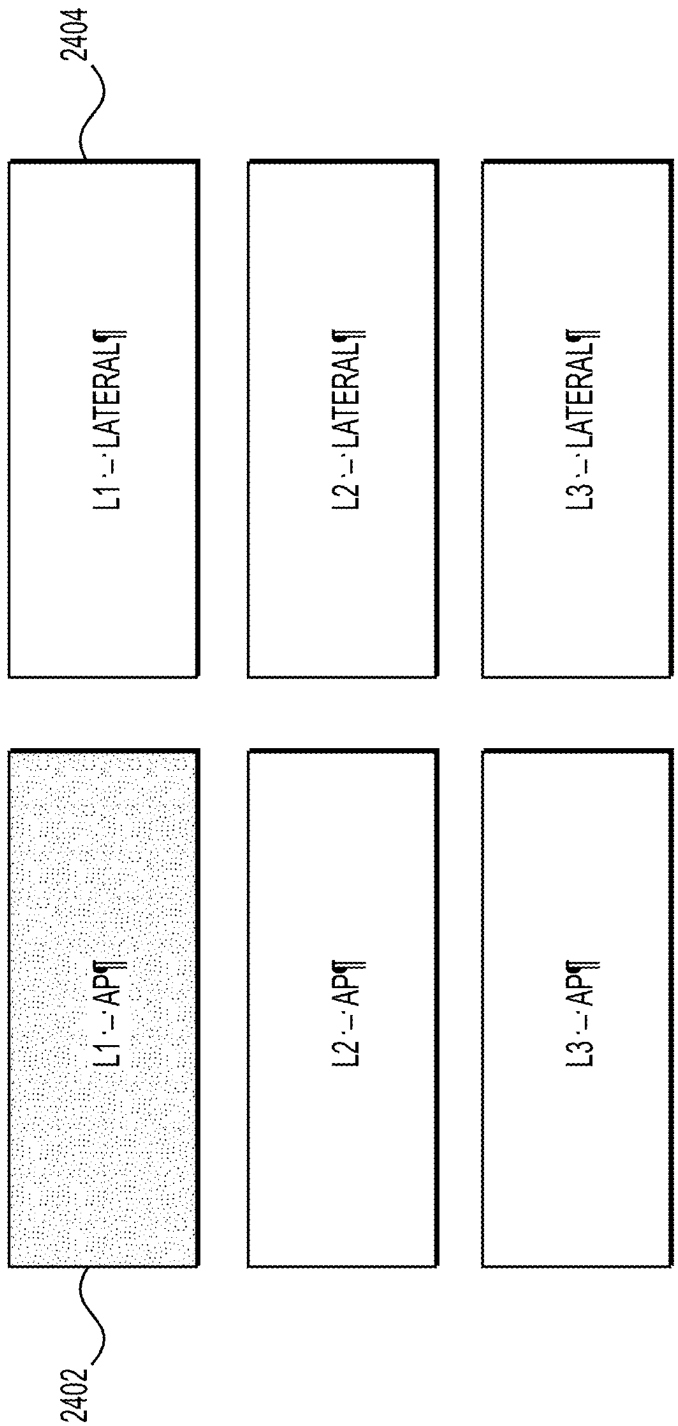
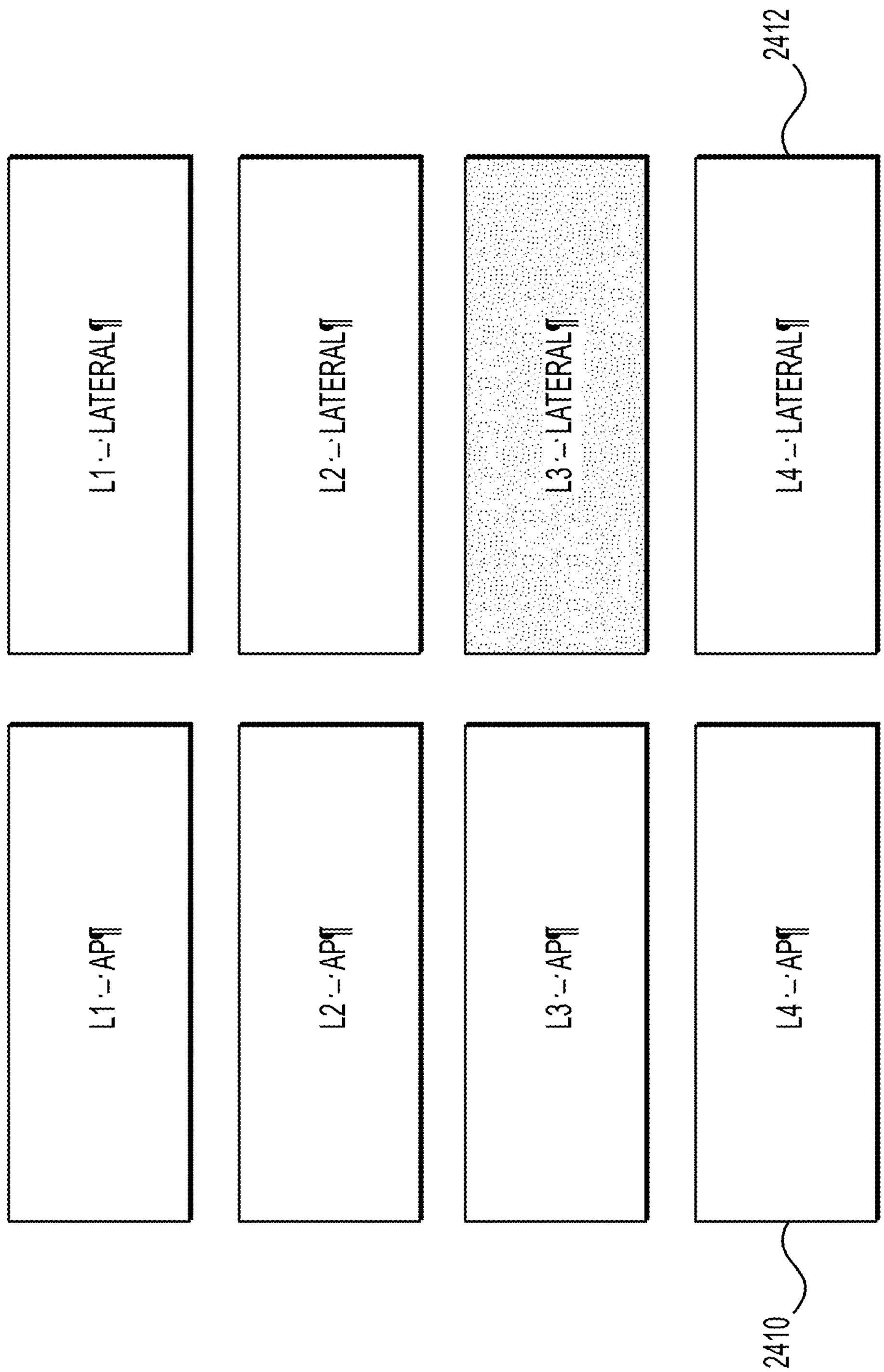


FIG. 24A



2412
L1 - LATERAL
L2 - LATERAL
L3 - LATERAL
L4 - LATERAL
L1 - AP
L2 - AP
L3 - AP
L4 - AP
2410

FIG. 24B

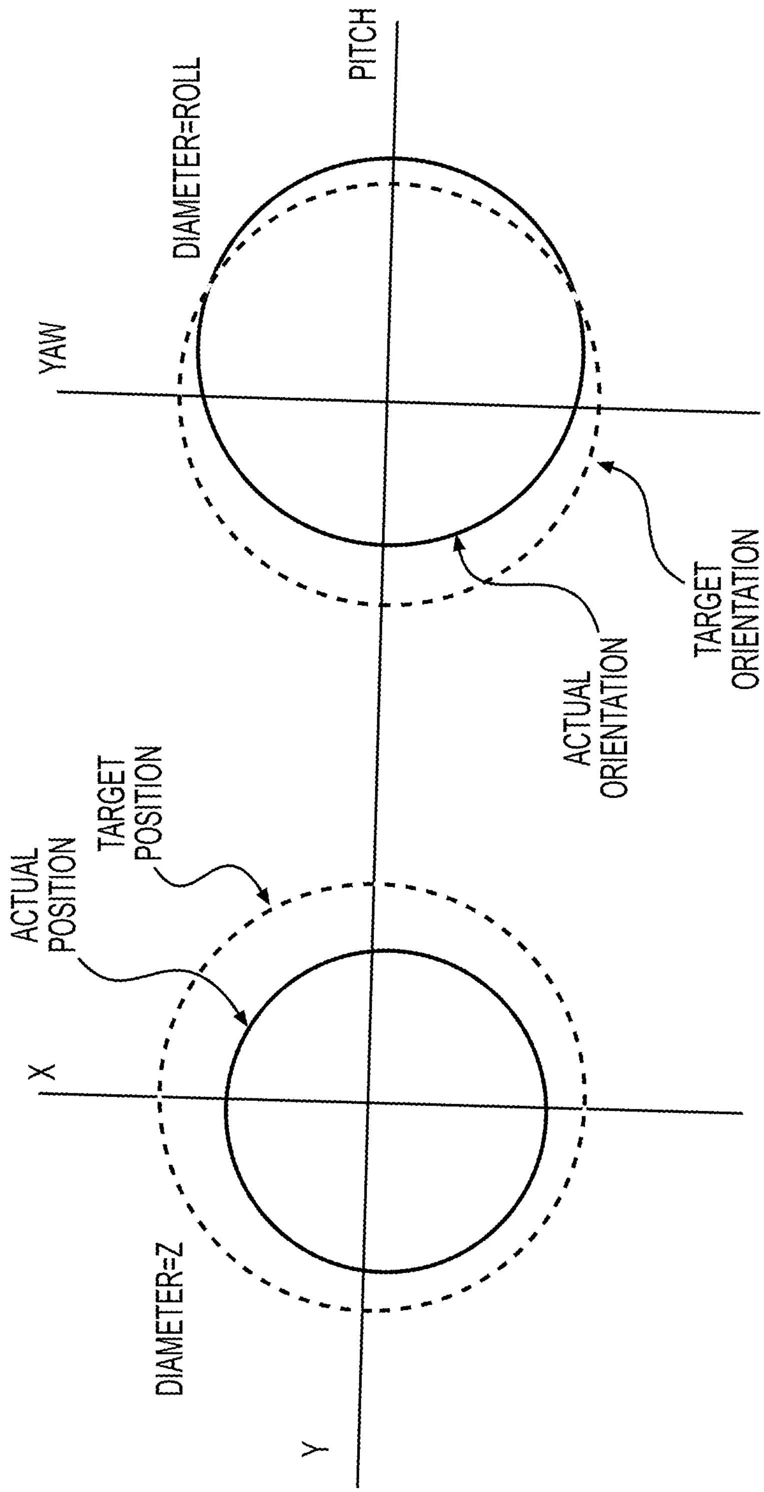
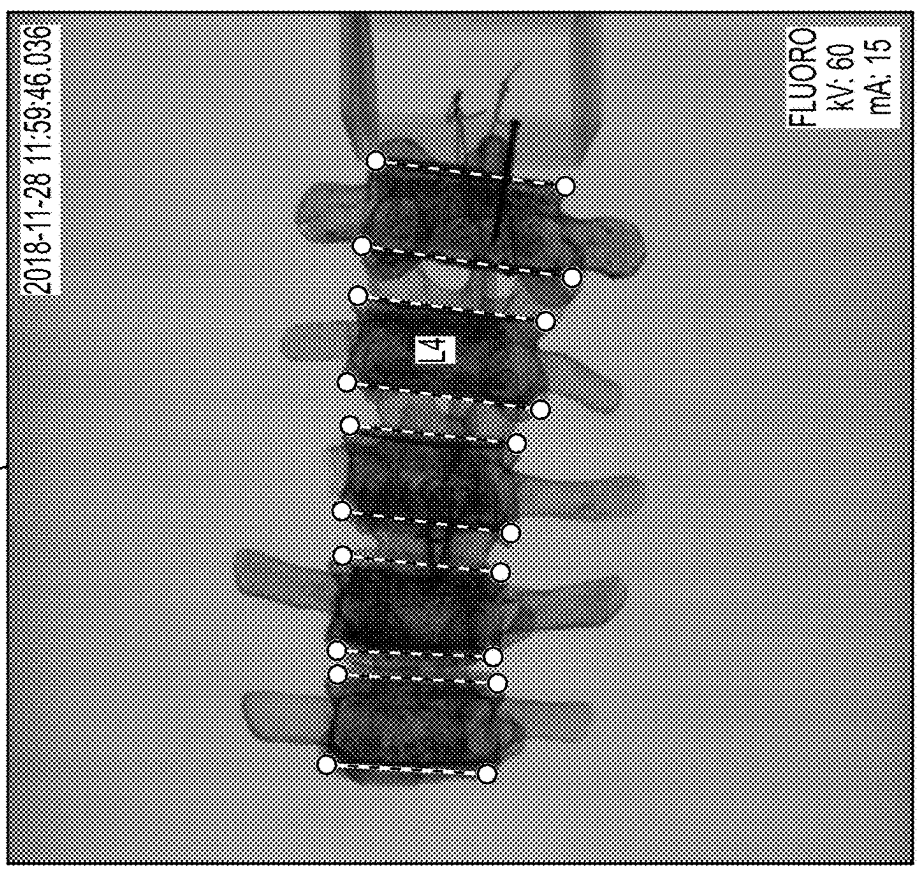

PITCH
DIAMETER=ROLL
YAW
ACTUAL ORIENTATION
TARGET ORIENTATION
ACTUAL POSITION
TARGET POSITION
X
DIAMETER=Z
Y

FIG. 25

2602



2604

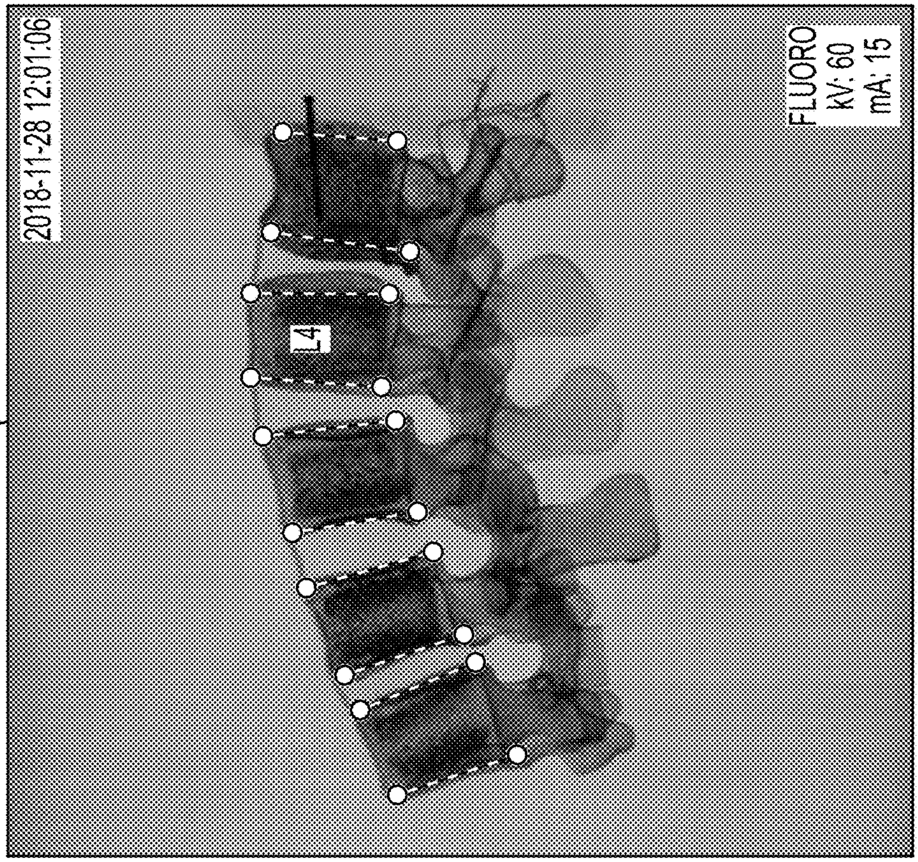


FIG. 26

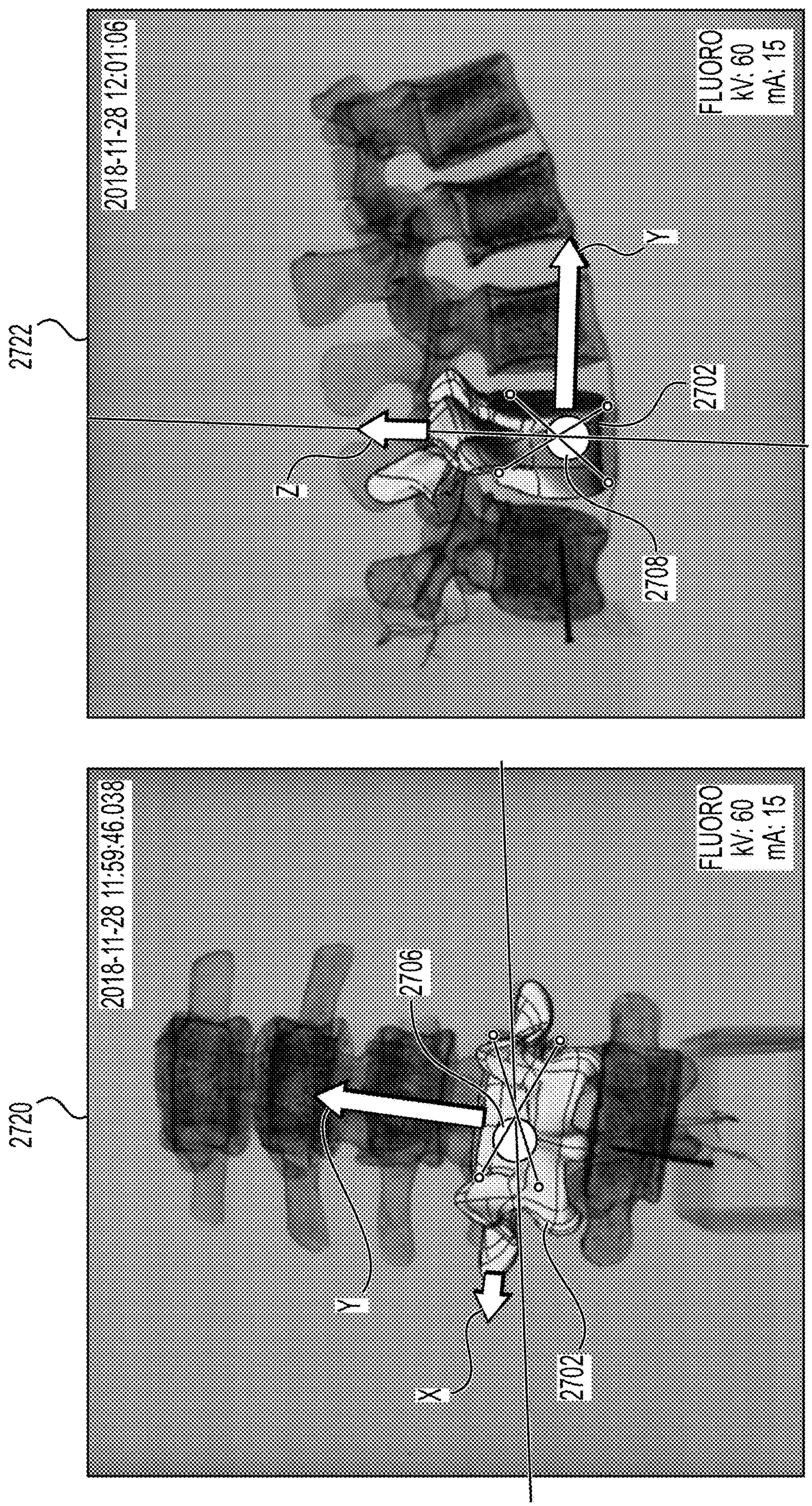

2722
2018-11-28 12:01:06
FLUORO
kV: 60
mA: 15
Y
Z
2702
2708
2720
2018-11-28 11:59:46.038
FLUORO
kV: 60
mA: 15
2706
Y
X
2702

FIG. 27A

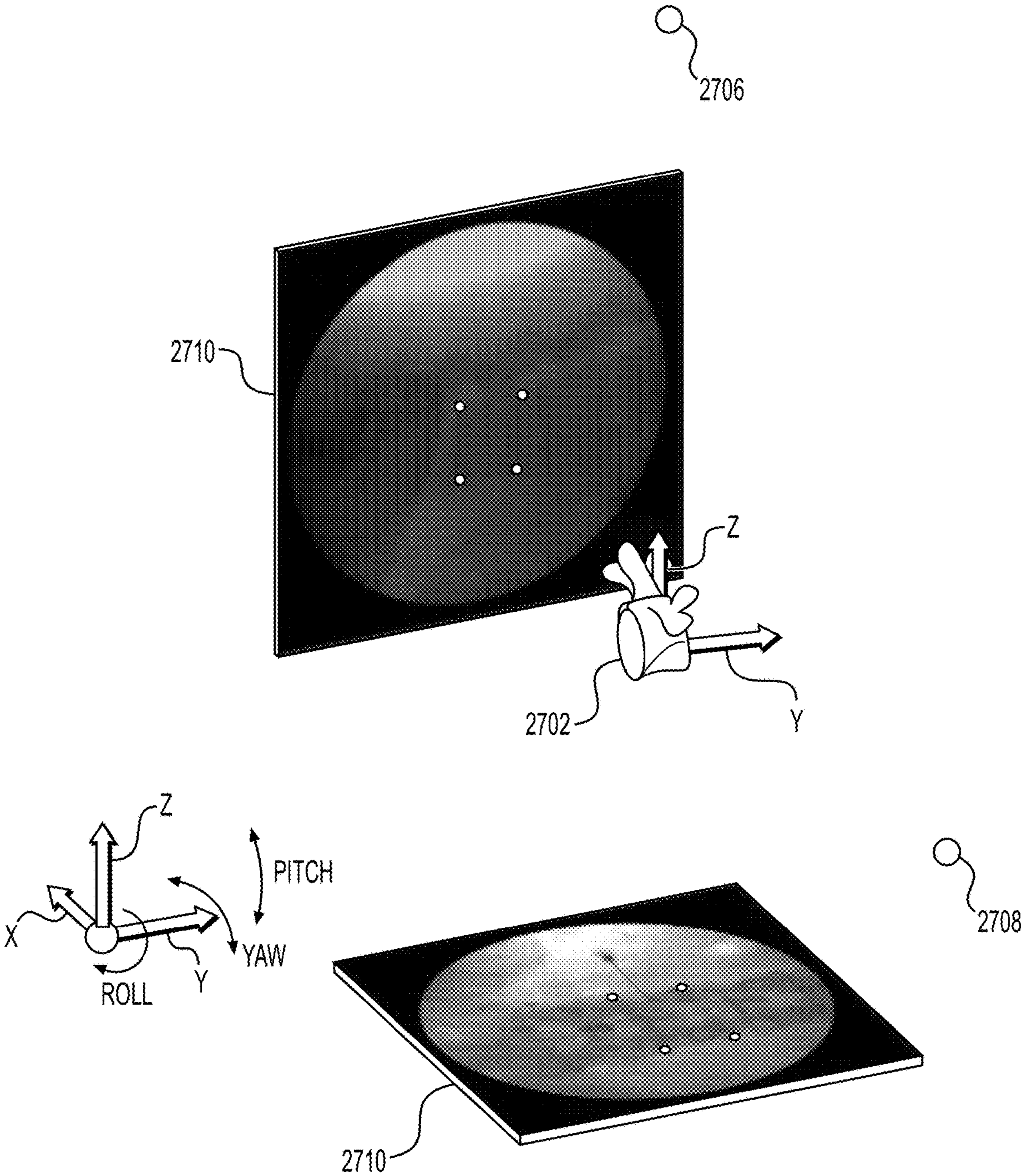


FIG. 27B

SYSTEM AND METHOD OF DETERMINING OPTIMAL 3-DIMENSIONAL POSITION AND ORIENTATION OF IMAGING DEVICE FOR IMAGING PATIENT BONES

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Application No. 62/977,435, filed Feb. 17, 2020 and U.S. Provisional Application No. 62/990,064, filed Mar. 16, 2020, all of which are incorporated herein by reference in their entirety for all purposes.

TECHNICAL FIELD

The present disclosure relates to surgical imaging systems, and in particular, system for determining 3-dimensional position and orientation of a 2-dimensional imaging device for a robot assisted surgery.

BACKGROUND

Surgeons use imaging of the spine to assist with accurate placement of implants such as interbody spacers and pedicle screws. Spine procedures can be open surgery where the surgeon has good visibility of the spine due to a larger incision or minimally invasive surgery (MIS) where the surgeon has no visibility of the spine due to very small incisions. As a spine procedure moves toward robot assisted or at least navigation assisted MIS, more x-ray images are needed to accurately place spinal implants. Today, C-arms are the main surgical imaging systems in use which produce 2-dimensional (2D) x-ray images (alternatively and interchangeably called “fluoro” images). A typical fluoro image is an Anterior-Posterior (AP) image (see **1902** in FIG. **19**A). Given that depth cannot be determined by one 2D image, a second image which is 90-degrees offset from the AP image is taken; this is called a lateral image **1904** as seen in FIG. **19**B.

It is important to align the C-arm properly with the spinal vertebral body so the surgeon can align the implant with the anatomy. FIG. **20** illustrates an oblique improperly aligned lateral image of the lumbar spine. With such a misaligned image, it would be very difficult to accurately place an implant even by an experienced surgeon.

To take optimal images, a radiological technician (RT) iteratively positions the C-arm or patient bed for each of the AP image and lateral image. After each position adjustment, a fluoro image is taken and examined. For an experienced RT, the number of shots to acquire a good AP or lateral image may be three. For an inexperienced RT, the number is greater than 10. Since each vertebral level requires two images (AP and lateral), a spine case involving three vertebral levels would need at least six fluoro shots for each level (three shots for AP and three shots for lateral) by an experienced RT, resulting in a total number of fluoro shots at 18. For an inexperienced RT, the total number of fluoro shots for the same three vertebral procedure may be higher than 60 (10 shots for AP and 10 shots for lateral).

As can be appreciated, a typical spine procedure exposes the patient, RT and surgeon to a large amount of radiation. Moreover, the large number of fluoro shots during surgery slows down the surgical procedure and creates substantially higher risk to the patients as well as higher costs.

Therefore, it would be desirable to provide a system and method for reducing the number of fluoro shots required for a spine surgery.

SUMMARY OF THE INVENTION

To meet this and other needs, devices, systems, and methods for determining the optimal 3-dimensional position and orientation of an imaging arm of an imaging device is provided.

According to one aspect of the present invention, a method of determining the imaging arm’s optimal 3-dimensional position and orientation for taking images of a vertebral body. According to the method, test images of the vertebral body are taken by the user and are received by the imaging device. The test images include an x-ray image of the vertebral body and a second x-ray image at a different angle than the first x-ray image. Typically, the two images would be taken approximately 90 degrees from each other. From the test images, vertebral bodies of interest are identified either by the user or a computer. The vertebral bodies are then segmented by the computer with or without help from the user. A 3-dimensional model of the vertebral body is then aligned against the corresponding vertebral body in the test images. Based on the alignment, a 3-dimensional position and orientation of the imaging arm for taking optimal AP and lateral x-ray images are determined based on the aligned 3-dimensional model.

By having the computer determine the optimal position and orientation of the imaging arm of the imaging device, the present method eliminates the need to repeatedly take fluoro shots manually to find the optimum images.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is an overhead view of a potential arrangement for locations of the robotic system, patient, surgeon, and other medical personnel during a surgical procedure;

FIG. **2** illustrates the robotic system including positioning of the surgical robot and the camera relative to the patient according to one embodiment;

FIG. **3** illustrates a surgical robotic system in accordance with an exemplary embodiment;

FIG. **4** illustrates a portion of a surgical robot in accordance with an exemplary embodiment;

FIG. **5** illustrates a block diagram of a surgical robot in accordance with an exemplary embodiment;

FIG. **6** illustrates a surgical robot in accordance with an exemplary embodiment;

FIGS. **7**A-**7**C illustrate an end-effector in accordance with an exemplary embodiment;

FIG. **8** illustrates a surgical instrument and the end-effector, before and after, inserting the surgical instrument into the guide tube of the end-effector according to one embodiment;

FIGS. **9**A-**9**C illustrate portions of an end-effector and robot arm in accordance with an exemplary embodiment;

FIG. **10** illustrates a dynamic reference array, an imaging array, and other components in accordance with an exemplary embodiment;

FIG. **11** illustrates a method of registration in accordance with an exemplary embodiment;

FIG. **12**A-**12**B illustrate embodiments of imaging devices according to exemplary embodiments;

FIG. **13**A illustrates a portion of a robot including the robot arm and an end-effector in accordance with an exemplary embodiment;

FIG. **13**B is a close-up view of the end-effector, with a plurality of tracking markers rigidly affixed thereon, shown in FIG. **13**A;

FIG. **13**C is a tool or instrument with a plurality of tracking markers rigidly affixed thereon according to one embodiment;

FIG. **14**A is an alternative version of an end-effector with moveable tracking markers in a first configuration;

FIG. **14**B is the end-effector shown in FIG. **14**A with the moveable tracking markers in a second configuration;

FIG. **14**C shows the template of tracking markers in the first configuration from FIG. **14**A;

FIG. **14**D shows the template of tracking markers in the second configuration from FIG. **14**B;

FIG. **15**A shows an alternative version of the end-effector having only a single tracking marker affixed thereto;

FIG. **15**B shows the end-effector of FIG. **15**A with an instrument disposed through the guide tube;

FIG. **15**C shows the end-effector of FIG. **15**A with the instrument in two different positions, and the resulting logic to determine if the instrument is positioned within the guide tube or outside of the guide tube;

FIG. **15**D shows the end-effector of FIG. **15**A with the instrument in the guide tube at two different frames and its relative distance to the single tracking marker on the guide tube;

FIG. **15**E shows the end-effector of FIG. **15**A relative to a coordinate system;

FIG. **16** is a block diagram of a method for navigating and moving the end-effector of the robot to a desired target trajectory;

FIGS. **17**A-**17**B depict an instrument for inserting an expandable implant having fixed and moveable tracking markers in contracted and expanded positions, respectively;

FIGS. **18**A-**18**B depict an instrument for inserting an articulating implant having fixed and moveable tracking markers in insertion and angled positions, respectively;

FIGS. **19**A and **19**B illustrate typical AP and lateral images of a spine;

FIG. **20** illustrates a misaligned lateral image of the spine;

FIG. **21** illustrates a method of determining the 3-dimensional position of an imaging arm of an imaging device for taking optimal images of a vertebral body according to one aspect of the present invention;

FIG. **22** illustrates a calibration ring having tracking markers according to one aspect of the present invention;

FIG. **23** is an example of an x-ray imaging device having an automatic positioning capability with respect to the 3D position and orientation of its C-arm according one aspect of the present invention;

FIG. **24**A is an example of a graphical user interface showing the available selection of images at different vertebral levels according to an aspect of the present invention;

FIG. **24**B is an example of a graphical user interface showing the additional available selection of images at adjacent vertebral levels according to an aspect of the present invention;

FIG. **25** is an example of a graphical user interface that guides a user to position the C-arm at an optimal position according to an aspect of the present invention;

FIG. **26** is a representation of a method of performing a segmentation of vertebral bodies according to an aspect of the present invention; and

FIGS. **27**A and **27**B graphically illustrate a method of aligning a 3D model of a selected vertebral body to the scanned AP and lateral images according to an aspect of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

It is to be understood that the present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the description herein or illustrated in the drawings. The teachings of the present disclosure may be used and practiced in other embodiments and practiced or carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of "including," "comprising," or "having" and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms "mounted," "connected," "supported," and "coupled" and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, "connected" and "coupled" are not restricted to physical or mechanical connections or couplings.

The following discussion is presented to enable a person skilled in the art to make and use embodiments of the present disclosure. Various modifications to the illustrated embodiments will be readily apparent to those skilled in the art, and the principles herein can be applied to other embodiments and applications without departing from embodiments of the present disclosure. Thus, the embodiments are not intended to be limited to embodiments shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein. The following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. The figures, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of the embodiments. Skilled artisans will recognize the examples provided herein have many useful alternatives and fall within the scope of the embodiments.

Turning now to the drawing, FIGS. **1** and **2** illustrate a surgical robot system **100** in accordance with an exemplary embodiment. Surgical robot system **100** may include, for example, a surgical robot **102**, one or more robot arms **104**, a base **106**, a display **110**, an end-effector **112**, for example, including a guide tube **114**, and one or more tracking markers **118**. The surgical robot system **100** may include a patient tracking device **116** also including one or more tracking markers **118**, which is adapted to be secured directly to the patient **210** (e.g., to the bone of the patient **210**). The surgical robot system **100** may also utilize a camera **200**, for example, positioned on a camera stand **202**. The camera stand **202** can have any suitable configuration to move, orient, and support the camera **200** in a desired position. The camera **200** may include any suitable camera or cameras, such as one or more infrared cameras (e.g., bifocal or stereophotogrammetric cameras), able to identify, for example, active and passive tracking markers **118** in a given measurement volume viewable from the perspective of the camera **200**. The camera **200** may scan the given measurement volume and detect the light that comes from the markers **118** in order to identify and determine the position of the markers **118** in three-dimensions. For example, active markers **118** may include infrared-emitting markers that are activated by an electrical signal (e.g., infrared light emitting diodes (LEDs)), and passive markers **118** may include retro-reflective markers that reflect infrared light (e.g., they reflect incoming IR radiation into the direction of the incoming light), for example, emitted by illuminators on the camera **200** or other suitable device.

FIGS. **1** and **2** illustrate a potential configuration for the placement of the surgical robot system **100** in an operating room environment. For example, the robot **102** may be positioned near or next to patient **210**. Although depicted near the head of the patient **210**, it will be appreciated that the robot **102** can be positioned at any suitable location near the patient **210** depending on the area of the patient **210** undergoing the operation. The camera **200** may be separated from the robot system **100** and positioned at the foot of patient **210**. This location allows the camera **200** to have a direct visual line of sight to the surgical field **208**. Again, it is contemplated that the camera **200** may be located at any suitable position having line of sight to the surgical field **208**. In the configuration shown, the surgeon **120** may be positioned across from the robot **102**, but is still able to manipulate the end-effector **112** and the display **110**. A surgical assistant **126** may be positioned across from the surgeon **120** again with access to both the end-effector **112** and the display **110**. If desired, the locations of the surgeon **120** and the assistant **126** may be reversed. The traditional areas for the anesthesiologist **122** and the nurse or scrub tech **124** remain unimpeded by the locations of the robot **102** and camera **200**.

With respect to the other components of the robot **102**, the display **110** can be attached to the surgical robot **102** and in other exemplary embodiments, display **110** can be detached from surgical robot **102**, either within a surgical room with the surgical robot **102**, or in a remote location. End-effector **112** may be coupled to the robot arm **104** and controlled by at least one motor. In exemplary embodiments, end-effector **112** can comprise a guide tube **114**, which is able to receive and orient a surgical instrument **608** (described further herein) used to perform surgery on the patient **210**. As used herein, the term “end-effector” is used interchangeably with the terms “end-effectuator” and “effectuator element.” Although generally shown with a guide tube **114**, it will be appreciated that the end-effector **112** may be replaced with any suitable instrumentation suitable for use in surgery. In some embodiments, end-effector **112** can comprise any known structure for effecting the movement of the surgical instrument **608** in a desired manner.

The surgical robot **102** is able to control the translation and orientation of the end-effector **112**. The robot **102** is able to move end-effector **112** along x-, y-, and z-axes, for example. The end-effector **112** can be configured for selective rotation about one or more of the x-, y-, and z-axis, and a Z Frame axis (such that one or more of the Euler Angles (e.g., roll, pitch, and/or yaw) associated with end-effector **112** can be selectively controlled). In some exemplary embodiments, selective control of the translation and orientation of end-effector **112** can permit performance of medical procedures with significantly improved accuracy compared to conventional robots that utilize, for example, a six degree of freedom robot arm comprising only rotational axes. For example, the surgical robot system **100** may be used to operate on patient **210**, and robot arm **104** can be positioned above the body of patient **210**, with end-effector **112** selectively angled relative to the z-axis toward the body of patient **210**.

In some exemplary embodiments, the position of the surgical instrument **608** can be dynamically updated so that surgical robot **102** can be aware of the location of the surgical instrument **608** at all times during the procedure. Consequently, in some exemplary embodiments, surgical robot **102** can move the surgical instrument **608** to the desired position quickly without any further assistance from a physician (unless the physician so desires). In some further embodiments, surgical robot **102** can be configured to correct the path of the surgical instrument **608** if the surgical instrument **608** strays from the selected, preplanned trajectory. In some exemplary embodiments, surgical robot **102** can be configured to permit stoppage, modification, and/or manual control of the movement of end-effector **112** and/or the surgical instrument **608**. Thus, in use, in exemplary embodiments, a physician or other user can operate the system **100**, and has the option to stop, modify, or manually control the autonomous movement of end-effector **112** and/or the surgical instrument **608**. Further details of surgical robot system **100** including the control and movement of a surgical instrument **608** by surgical robot **102** can be found in co-pending U.S. patent application Ser. No. 13/924,505, which is incorporated herein by reference in its entirety.

The robotic surgical system **100** can comprise one or more tracking markers **118** configured to track the movement of robot arm **104**, end-effector **112**, patient **210**, and/or the surgical instrument **608** in three dimensions. In exemplary embodiments, a plurality of tracking markers **118** can be mounted (or otherwise secured) thereon to an outer surface of the robot **102**, such as, for example and without limitation, on base **106** of robot **102**, on robot arm **104**, or on the end-effector **112**. In exemplary embodiments, at least one tracking marker **118** of the plurality of tracking markers **118** can be mounted or otherwise secured to the end-effector **112**. One or more tracking markers **118** can further be mounted (or otherwise secured) to the patient **210**. In exemplary embodiments, the plurality of tracking markers **118** can be positioned on the patient **210** spaced apart from the surgical field **208** to reduce the likelihood of being obscured by the surgeon, surgical tools, or other parts of the robot **102**. Further, one or more tracking markers **118** can be further mounted (or otherwise secured) to the surgical tools **608** (e.g., a screw driver, dilator, implant inserter, or the like). Thus, the tracking markers **118** enable each of the marked objects (e.g., the end-effector **112**, the patient **210**, and the surgical tools **608**) to be tracked by the robot **102**. In exemplary embodiments, system **100** can use tracking information collected from each of the marked objects to calculate the orientation and location, for example, of the end-effector **112**, the surgical instrument **608** (e.g., positioned in the tube **114** of the end-effector **112**), and the relative position of the patient **210**.

In exemplary embodiments, one or more of markers **118** may be optical markers. In some embodiments, the positioning of one or more tracking markers **118** on end-effector **112** can maximize the accuracy of the positional measurements by serving to check or verify the position of end-effector **112**. Further details of surgical robot system **100** including the control, movement and tracking of surgical robot **102** and of a surgical instrument **608** can be found in co-pending U.S. patent application Ser. No. 13/924,505, which is incorporated herein by reference in its entirety.

Exemplary embodiments include one or more markers **118** coupled to the surgical instrument **608**. In exemplary embodiments, these markers **118**, for example, coupled to the patient **210** and surgical instruments **608**, as well as markers **118** coupled to the end-effector **112** of the robot **102** can comprise conventional infrared light-emitting diodes (LEDs) or an Optotrak® diode capable of being tracked using a commercially available infrared optical tracking system such as Optotrak®. Optotrak® is a registered trademark of Northern Digital Inc., Waterloo, Ontario, Canada. In other embodiments, markers **118** can comprise conventional reflective spheres capable of being tracked using a commercially available optical tracking system such as Polaris Spectra. Polaris Spectra is also a registered trademark of Northern Digital, Inc. In an exemplary embodiment, the markers **118** coupled to the end-effector **112** are active markers which comprise infrared light-emitting diodes which may be turned on and off, and the markers **118** coupled to the patient **210** and the surgical instruments **608** comprise passive reflective spheres.

In exemplary embodiments, light emitted from and/or reflected by markers **118** can be detected by camera **200** and can be used to monitor the location and movement of the marked objects. In alternative embodiments, markers **118** can comprise a radio-frequency and/or electromagnetic reflector or transceiver and the camera **200** can include or be replaced by a radio-frequency and/or electromagnetic transceiver.

Similar to surgical robot system **100**, FIG. **3** illustrates a surgical robot system **300** and camera stand **302**, in a docked configuration, consistent with an exemplary embodiment of the present disclosure. Surgical robot system **300** may comprise a robot **301** including a display **304**, upper arm **306**, lower arm **308**, end-effector **310**, vertical column **312**, casters **314**, cabinet **316**, tablet drawer **318**, connector panel **320**, control panel **322**, and ring of information **324**. Camera stand **302** may comprise camera **326**. These components are described in greater with respect to FIG. **5**. FIG. **3** illustrates the surgical robot system **300** in a docked configuration where the camera stand **302** is nested with the robot **301**, for example, when not in use. It will be appreciated by those skilled in the art that the camera **326** and robot **301** may be separated from one another and positioned at any appropriate location during the surgical procedure, for example, as shown in FIGS. **1** and **2**.

FIG. **4** illustrates a base **400** consistent with an exemplary embodiment of the present disclosure. Base **400** may be a portion of surgical robot system **300** and comprise cabinet **316**. Cabinet **316** may house certain components of surgical robot system **300** including but not limited to a battery **402**, a power distribution module **404**, a platform interface board module **406**, a computer **408**, a handle **412**, and a tablet drawer **414**. The connections and relationship between these components is described in greater detail with respect to FIG. **5**.

FIG. **5** illustrates a block diagram of certain components of an exemplary embodiment of surgical robot system **300**. Surgical robot system **300** may comprise platform subsystem **502**, computer subsystem **504**, motion control subsystem **506**, and tracking subsystem **532**. Platform subsystem **502** may further comprise battery **402**, power distribution module **404**, platform interface board module **406**, and tablet charging station **534**. Computer subsystem **504** may further comprise computer **408**, display **304**, and speaker **536**. Motion control subsystem **506** may further comprise driver circuit **508**, motors **510**, **512**, **514**, **516**, **518**, stabilizers **520**, **522**, **524**, **526**, end-effector **310**, and controller **538**. Tracking subsystem **532** may further comprise position sensor **540** and camera converter **542**. System **300** may also comprise a foot pedal **544** and tablet **546**.

Input power is supplied to system **300** via a power source **548** which may be provided to power distribution module **404**. Power distribution module **404** receives input power and is configured to generate different power supply voltages that are provided to other modules, components, and subsystems of system **300**. Power distribution module **404** may be configured to provide different voltage supplies to platform interface module **406**, which may be provided to other components such as computer **408**, display **304**, speaker **536**, driver **508** to, for example, power motors **512**, **514**, **516**, **518** and end-effector **310**, motor **510**, ring **324**, camera converter **542**, and other components for system **300** for example, fans for cooling the electrical components within cabinet **316**.

Power distribution module **404** may also provide power to other components such as tablet charging station **534** that may be located within tablet drawer **318**. Tablet charging station **534** may be in wireless or wired communication with tablet **546** for charging table **546**. Tablet **546** may be used by a surgeon consistent with the present disclosure and described herein.

Power distribution module **404** may also be connected to battery **402**, which serves as temporary power source in the event that power distribution module **404** does not receive power from input power **548**. At other times, power distribution module **404** may serve to charge battery **402** if necessary.

Other components of platform subsystem **502** may also include connector panel **320**, control panel **322**, and ring **324**. Connector panel **320** may serve to connect different devices and components to system **300** and/or associated components and modules. Connector panel **320** may contain one or more ports that receive lines or connections from different components. For example, connector panel **320** may have a ground terminal port that may ground system **300** to other equipment, a port to connect foot pedal **544** to system **300**, a port to connect to tracking subsystem **532**, which may comprise position sensor **540**, camera converter **542**, and cameras **326** associated with camera stand **302**. [A PORT IN THE CONNECTOR PANEL **320** MAY ALSO CONNECT TO AN IMAGING DEVICE FOR RECEIVING SCANNED IMAGES AND FOR CONTROLLING THE LOCATION AND ORIENTATION OF THE C-ARM BASED ON THE OPTICAL/NAVIGATION MARKERS ATTACHED TO THE IMAGING DEVICE] Connector panel **320** may also include other ports to allow USB, Ethernet, HDMI communications to other components, such as computer **408**.

Control panel **322** may provide various buttons or indicators that control operation of system **300** and/or provide information regarding system **300**. For example, control panel **322** may include buttons to power on or off system **300**, lift or lower vertical column **312**, and lift or lower stabilizers **520**-**526** that may be designed to engage casters **314** to lock system **300** from physically moving. Other buttons may stop system **300** in the event of an emergency, which may remove all motor power and apply mechanical brakes to stop all motion from occurring. Control panel **322** may also have indicators notifying the user of certain system conditions such as a line power indicator or status of charge for battery **402**.

Ring **324** may be a visual indicator to notify the user of system **300** of different modes that system **300** is operating under and certain warnings to the user.

Computer subsystem **504** includes computer **408**, display **304**, and speaker **536**. Computer **504** includes an operating system and software to operate system **300**. Computer **504** may receive and process information from other components (for example, tracking subsystem **532**, platform subsystem **502**, and/or motion control subsystem **506**) in order to display information to the user. Further, computer subsystem **504** may also include speaker **536** to provide audio to the user.

Tracking subsystem **532** may include position sensor **504** and converter **542**. Tracking subsystem **532** may correspond to camera stand **302** including camera **326** as described with respect to FIG. **3**. Position sensor **504** may be camera **326**. Tracking subsystem may track the location of certain markers that are located on the different components of system **300** and/or instruments used by a user during a surgical procedure. This tracking may be conducted in a manner consistent with the present disclosure including the use of infrared technology that tracks the location of active or passive elements, such as LEDs or reflective markers, respectively. The location, orientation, and position of structures having these types of markers may be provided to computer **408** which may be shown to a user on display **304**. For example, a surgical instrument **608** having these types of markers and tracked in this manner (which may be referred to as a navigational space) may be shown to a user in relation to a three dimensional image of a patient's anatomical structure.

Motion control subsystem **506** may be configured to physically move vertical column **312**, upper arm **306**, lower arm **308**, or rotate end-effector **310**. The physical movement may be conducted through the use of one or more motors **510**-**518**. For example, motor **510** may be configured to vertically lift or lower vertical column **312**. Motor **512** may be configured to laterally move upper arm **308** around a point of engagement with vertical column **312** as shown in FIG. **3**. Motor **514** may be configured to laterally move lower arm **308** around a point of engagement with upper arm **308** as shown in FIG. **3**. Motors **516** and **518** may be configured to move end-effector **310** in a manner such that one may control the roll and one may control the tilt, thereby providing multiple angles that end-effector **310** may be moved. These movements may be achieved by controller **538** which may control these movements through load cells disposed on end-effector **310** and activated by a user engaging these load cells to move system **300** in a desired manner.

Moreover, system **300** may provide for automatic movement of vertical column **312**, upper arm **306**, and lower arm **308** through a user indicating on display **304** (which may be a touchscreen input device) the location of a surgical instrument or component on three dimensional image of the patient's anatomy on display **304**. The user may initiate this automatic movement by stepping on foot pedal **544** or some other input means.

FIG. **6** illustrates a surgical robot system **600** consistent with an exemplary embodiment. Surgical robot system **600** may comprise end-effector **602**, robot arm **604**, guide tube **606**, instrument **608**, and robot base **610**. Instrument tool **608** may be attached to a tracking array **612** including one or more tracking markers (such as markers **118**) and have an associated trajectory **614**. Trajectory **614** may represent a path of movement that instrument tool **608** is configured to travel once it is positioned through or secured in guide tube **606**, for example, a path of insertion of instrument tool **608** into a patient. In an exemplary operation, robot base **610** may be configured to be in electronic communication with robot arm **604** and end-effector **602** so that surgical robot system **600** may assist a user (for example, a surgeon) in operating on the patient **210**. Surgical robot system **600** may be consistent with previously described surgical robot system **100** and **300**.

A tracking array **612** may be mounted on instrument **608** to monitor the location and orientation of instrument tool **608**. The tracking array **612** may be attached to an instrument **608** and may comprise tracking markers **804**. As best seen in FIG. **8**, tracking markers **804** may be, for example, light emitting diodes and/or other types of reflective markers (e.g., markers **118** as described elsewhere herein). The tracking devices may be one or more line of sight devices associated with the surgical robot system. As an example, the tracking devices may be one or more cameras **200**, **326** associated with the surgical robot system **100**, **300** and may also track tracking array **612** for a defined domain or relative orientations of the instrument **608** in relation to the robot arm **604**, the robot base **610**, end-effector **602**, and/or the patient **210**. The tracking devices may be consistent with those structures described in connection with camera stand **302** and tracking subsystem **532**.

FIGS. **7**A, **7**B, and **7**C illustrate a top view, front view, and side view, respectively, of end-effector **602** consistent with an exemplary embodiment. End-effector **602** may comprise one or more tracking markers **702**. Tracking markers **702** may be light emitting diodes or other types of active and passive markers, such as tracking markers **118** that have been previously described. In an exemplary embodiment, the tracking markers **702** are active infrared-emitting markers that are activated by an electrical signal (e.g., infrared light emitting diodes (LEDs)). Thus, tracking markers **702** may be activated such that the infrared markers **702** are visible to the camera **200**, **326** or may be deactivated such that the infrared markers **702** are not visible to the camera **200**, **326**. Thus, when the markers **702** are active, the end-effector **602** may be controlled by the system **100**, **300**, **600**, and when the markers **702** are deactivated, the end-effector **602** may be locked in position and unable to be moved by the system **100**, **300**, **600**.

Markers **702** may be disposed on or within end-effector **602** in a manner such that the markers **702** are visible by one or more cameras **200**, **326** or other tracking devices associated with the surgical robot system **100**, **300**, **600**. The camera **200**, **326** or other tracking devices may track end-effector **602** as it moves to different positions and viewing angles by following the movement of tracking markers **702**. The location of markers **702** and/or end-effector **602** may be shown on a display **110**, **304** associated with the surgical robot system **100**, **300**, **600**, for example, display **110** as shown in FIG. **2** and/or display **304** shown in FIG. **3**. This display **110**, **304** may allow a user to ensure that end-effector **602** is in a desirable position in relation to robot arm **604**, robot base **610**, the patient **210**, and/or the user.

For example, as shown in FIG. **7**A, markers **702** may be placed around the surface of end-effector **602** so that a tracking device placed away from the surgical field **208** and facing toward the robot **102**, **301** and the camera **200**, **326** is able to view at least 3 of the markers **702** through a range of common orientations of the end-effector **602** relative to the tracking device **100**, **300**, **600**. For example, distribution of markers **702** in this way allows end-effector **602** to be monitored by the tracking devices when end-effector **602** is translated and rotated in the surgical field **208**.

In addition, in exemplary embodiments, end-effector **602** may be equipped with infrared (IR) receivers that can detect when an external camera **200**, **326** is getting ready to read markers **702**. Upon this detection, end-effector **602** may then illuminate markers **702**. The detection by the IR receivers that the external camera**200**, **326** is ready to read markers **702** may signal the need to synchronize a duty cycle of markers **702**, which may be light emitting diodes, to an external camera**200**, **326**. This may also allow for lower power consumption by the robotic system as a whole, whereby markers **702** would only be illuminated at the appropriate time instead of being illuminated continuously. Further, in exemplary embodiments, markers **702** may be powered off to prevent interference with other navigation tools, such as different types of surgical instruments **608**.

FIG. **8** depicts one type of surgical instrument **608** including a tracking array **612** and tracking markers **804**. Tracking markers **804** may be of any type described herein including but not limited to light emitting diodes or reflective spheres. Markers **804** are monitored by tracking devices associated with the surgical robot system **100**, **300**, **600** and may be one or more of the line of sight cameras **200**, **326**. The cameras **200**, **326** may track the location of instrument **608** based on the position and orientation of tracking array **612** and markers **804**. A user, such as a surgeon **120**, may orient instrument **608** in a manner so that tracking array **612** and markers **804** are sufficiently recognized by the tracking device or camera **200**, **326** to display instrument **608** and markers **804** on, for example, display **110** of the exemplary surgical robot system.

The manner in which a surgeon **120** may place instrument **608** into guide tube **606** of the end-effector **602** and adjust the instrument **608** is evident in FIG. **8**. The hollow tube or guide tube **114**, **606** of the end-effector **112**, **310**, **602** is sized and configured to receive at least a portion of the surgical instrument **608**. The guide tube **114**, **606** is configured to be oriented by the robot arm **104** such that insertion and trajectory for the surgical instrument **608** is able to reach a desired anatomical target within or upon the body of the patient **210**. The surgical instrument **608** may include at least a portion of a generally cylindrical instrument. Although a screw driver is exemplified as the surgical tool **608**, it will be appreciated that any suitable surgical tool **608** may be positioned by the end-effector **602**. By way of example, the surgical instrument **608** may include one or more of a guide wire, cannula, a retractor, a drill, a reamer, a screw driver, an insertion tool, a removal tool, or the like. Although the hollow tube **114**, **606** is generally shown as having a cylindrical configuration, it will be appreciated by those of skill in the art that the guide tube **114**, **606** may have any suitable shape, size and configuration desired to accommodate the surgical instrument **608** and access the surgical site.

FIGS. **9**A-**9**C illustrate end-effector **602** and a portion of robot arm **604** consistent with an exemplary embodiment. End-effector **602** may further comprise body **1202** and clamp **1204**. Clamp **1204** may comprise handle **1206**, balls **1208**, spring **1210**, and lip **1212**. Robot arm **604** may further comprise depressions **1214**, mounting plate **1216**, lip **1218**, and magnets **1220**.

End-effector **602** may mechanically interface and/or engage with the surgical robot system and robot arm **604** through one or more couplings. For example, end-effector **602** may engage with robot arm **604** through a locating coupling and/or a reinforcing coupling. Through these couplings, end-effector **602** may fasten with robot arm **604** outside a flexible and sterile barrier. In an exemplary embodiment, the locating coupling may be a magnetically kinematic mount and the reinforcing coupling may be a five bar over center clamping linkage.

With respect to the locating coupling, robot arm **604** may comprise mounting plate **1216**, which may be non-magnetic material, one or more depressions **1214**, lip **1218**, and magnets **1220**. Magnet **1220** is mounted below each of depressions **1214**. Portions of clamp **1204** may comprise magnetic material and be attracted by one or more magnets **1220**. Through the magnetic attraction of clamp **1204** and robot arm **604**, balls **1208** become seated into respective depressions **1214**. For example, balls **1208** as shown in FIG. **9**B would be seated in depressions **1214** as shown in FIG. **9**A. This seating may be considered a magnetically-assisted kinematic coupling. Magnets **1220** may be configured to be strong enough to support the entire weight of end-effector **602** regardless of the orientation of end-effector **602**. The locating coupling may be any style of kinematic mount that uniquely restrains six degrees of freedom.

With respect to the reinforcing coupling, portions of clamp **1204** may be configured to be a fixed ground link and as such clamp **1204** may serve as a five bar linkage. Closing clamp handle **1206** may fasten end-effector **602** to robot arm **604** as lip **1212** and lip **1218** engage clamp **1204** in a manner to secure end-effector **602** and robot arm **604**. When clamp handle **1206** is closed, spring **1210** may be stretched or stressed while clamp **1204** is in a locked position. The locked position may be a position that provides for linkage past center. Because of a closed position that is past center, the linkage will not open absent a force applied to clamp handle **1206** to release clamp **1204**. Thus, in a locked position end-effector **602** may be robustly secured to robot arm **604**.

Spring **1210** may be a curved beam in tension. Spring **1210** may be comprised of a material that exhibits high stiffness and high yield strain such as virgin PEEK (poly-ether-ether-ketone). The linkage between end-effector **602** and robot arm **604** may provide for a sterile barrier between end-effector **602** and robot arm **604** without impeding fastening of the two couplings.

The reinforcing coupling may be a linkage with multiple spring members. The reinforcing coupling may latch with a cam or friction based mechanism. The reinforcing coupling may also be a sufficiently powerful electromagnet that will support fastening end-effector **102** to robot arm **604**. The reinforcing coupling may be a multi-piece collar completely separate from either end-effector **602** and/or robot arm **604** that slips over an interface between end-effector **602** and robot arm **604** and tightens with a screw mechanism, an over center linkage, or a cam mechanism.

Referring to FIGS. **10** and **11**, prior to or during a surgical procedure, certain registration procedures may be conducted in order to track objects and a target anatomical structure of the patient **210** both in a navigation space and an image space. In order to conduct such registration, a registration system **1400** may be used as illustrated in FIG. **10**.

In order to track the position of the patient **210**, a patient tracking device **116** may include a patient fixation instrument **1402** to be secured to a rigid anatomical structure of the patient **210** and a dynamic reference base (DRB) **1404** may be securely attached to the patient fixation instrument **1402**. For example, patient fixation instrument **1402** may be inserted into opening **1406** of dynamic reference base **1404**. Dynamic reference base **1404** may contain markers **1408** that are visible to tracking devices, such as tracking subsystem **532**. These markers **1408** may be optical markers or reflective spheres, such as tracking markers **118**, as previously discussed herein.

Patient fixation instrument **1402** is attached to a rigid anatomy of the patient **210** and may remain attached throughout the surgical procedure. In an exemplary embodiment, patient fixation instrument **1402** is attached to a rigid area of the patient **210**, for example, a bone that is located away from the targeted anatomical structure subject to the surgical procedure. In order to track the targeted anatomical structure, dynamic reference base **1404** is associated with the targeted anatomical structure through the use of a registration fixture that is temporarily placed on or near the targeted anatomical structure in order to register the dynamic reference base **1404** with the location of the targeted anatomical structure.

A registration fixture **1410** is attached to patient fixation instrument **1402** through the use of a pivot arm **1412**. Pivot arm **1412** is attached to patient fixation instrument **1402** by inserting patient fixation instrument **1402** through an opening **1414** of registration fixture **1410**. Pivot arm **1412** is attached to registration fixture **1410** by, for example, inserting a knob **1416** through an opening **1418** of pivot arm **1412**.

Using pivot arm **1412**, registration fixture **1410** may be placed over the targeted anatomical structure and its location may be determined in an image space and navigation space using tracking markers **1420** and/or fiducials **1422** on registration fixture **1410**. Registration fixture **1410** may contain a collection of markers **1420** that are visible in a navigational space (for example, markers **1420** may be detectable by tracking subsystem **532**). Tracking markers **1420** may be optical markers visible in infrared light as previously described herein. Registration fixture **1410** may also contain a collection of fiducials **1422**, for example, such as bearing balls, that are visible in an imaging space (for example, a three dimension CT image). As described in greater detail with respect to FIG. **11**, using registration fixture **1410**, the targeted anatomical structure may be associated with dynamic reference base **1404** thereby allowing depictions of objects in the navigational space to be overlaid on images of the anatomical structure. Dynamic reference base **1404**, located at a position away from the targeted anatomical structure, may become a reference point thereby allowing removal of registration fixture **1410** and/or pivot arm **1412** from the surgical area.

FIG. **11** provides an exemplary method **1500** for registration consistent with the present disclosure. Method **1500** begins at step **1502** wherein a graphical representation (or image(s)) of the targeted anatomical structure may be imported into system **100**, **300** **600**, for example computer **408**. The graphical representation may be three dimensional CT or a fluoroscope scan of the targeted anatomical structure of the patient **210** which includes registration fixture **1410** and a detectable imaging pattern of fiducials **1420**.

At step **1504**, an imaging pattern of fiducials **1420** is detected and registered in the imaging space and stored in computer **408**. Optionally, at this time at step **1506**, a graphical representation of the registration fixture **1410** may be overlaid on the images of the targeted anatomical structure.

At step **1508**, a navigational pattern of registration fixture **1410** is detected and registered by recognizing markers **1420**. Markers **1420** may be optical markers that are recognized in the navigation space through infrared light by tracking subsystem **532** via position sensor **540**. Thus, the location, orientation, and other information of the targeted anatomical structure is registered in the navigation space. Therefore, registration fixture **1410** may be recognized in both the image space through the use of fiducials **1422** and the navigation space through the use of markers **1420**. At step **1510**, the registration of registration fixture **1410** in the image space is transferred to the navigation space. This transferal is done, for example, by using the relative position of the imaging pattern of fiducials **1422** compared to the position of the navigation pattern of markers **1420**.

At step **1512**, registration of the navigation space of registration fixture **1410** (having been registered with the image space) is further transferred to the navigation space of dynamic registration array **1404** attached to patient fixture instrument **1402**. Thus, registration fixture **1410** may be removed and dynamic reference base **1404** may be used to track the targeted anatomical structure in both the navigation and image space because the navigation space is associated with the image space.

At steps **1514** and **1516**, the navigation space may be overlaid on the image space and objects with markers visible in the navigation space (for example, surgical instruments **608** with optical markers **804**). The objects may be tracked through graphical representations of the surgical instrument **608** on the images of the targeted anatomical structure.

FIGS. **12**A-**12**B illustrate imaging devices **1304** that may be used in conjunction with robot systems **100**, **300**, **600** to acquire pre-operative, intra-operative, post-operative, and/or real-time image data of patient **210**. Any appropriate subject matter may be imaged for any appropriate procedure using the imaging system **1304**. The imaging system **1304** may be any imaging device such as imaging device **1306** and/or a C-arm **1308** device. It may be desirable to take x-rays of patient **210** from a number of different positions, without the need for frequent manual repositioning of patient **210** which may be required in an x-ray system. As illustrated in FIG. **12**A, the imaging system **1304** may be in the form of a C-arm **1308** that includes an elongated C-shaped member terminating in opposing distal ends **1312** of the "C" shape. C-shaped member **1130** may further comprise an x-ray source **1314** and an image receptor **1316**. The space within C-arm **1308** of the arm may provide room for the physician to attend to the patient substantially free of interference from x-ray support structure **1318**. As illustrated in FIG. **12**B, the imaging system may include imaging device **1306** having a gantry housing **1324** attached to a support structure imaging device support structure **1328**, such as a wheeled mobile cart **1330** with wheels **1332**, which may enclose an image capturing portion, not illustrated. The image capturing portion may include an x-ray source and/or emission portion and an x-ray receiving and/or image receiving portion, which may be disposed about one hundred and eighty degrees from each other and mounted on a rotor (not illustrated) relative to a track of the image capturing portion. The image capturing portion may be operable to rotate three hundred and sixty degrees during image acquisition. The image capturing portion may rotate around a central point and/or axis, allowing image data of patient **210** to be acquired from multiple directions or in multiple planes. Although certain imaging systems **1304** are exemplified herein, it will be appreciated that any suitable imaging system may be selected by one of ordinary skill in the art.

Turning now to FIGS. **13**A-**13**C, the surgical robot system **100**, **300**, **600** relies on accurate positioning of the end-effector **112**, **602**, surgical instruments **608**, and/or the patient **210** (e.g., patient tracking device **116**) relative to the desired surgical area. In the embodiments shown in FIGS. **13**A-**13**C, the tracking markers **118**, **804** are rigidly attached to a portion of the instrument **608** and/or end-effector **112**.

FIG. **13**A depicts part of the surgical robot system **100** with the robot **102** including base **106**, robot arm **104**, and end-effector **112**. The other elements, not illustrated, such as the display, cameras, etc. may also be present as described herein. FIG. **13**B depicts a close-up view of the end-effector **112** with guide tube **114** and a plurality of tracking markers **118** rigidly affixed to the end-effector **112**. In this embodiment, the plurality of tracking markers **118** are attached to the guide tube **112**. FIG. **13**C depicts an instrument **608** (in this case, a probe **608**A) with a plurality of tracking markers **804** rigidly affixed to the instrument **608**. As described elsewhere herein, the instrument **608** could include any suitable surgical instrument, such as, but not limited to, guide wire, cannula, a retractor, a drill, a reamer, a screw driver, an insertion tool, a removal tool, or the like.

When tracking an instrument **608**, end-effector **112**, or other object to be tracked in 3D, an array of tracking markers

118, 804 may be rigidly attached to a portion of the tool 608 or end-effector 112. Preferably, the tracking markers 118, 804 are attached such that the markers 118, 804 are out of the way (e.g., not impeding the surgical operation, visibility, etc.). The markers 118, 804 may be affixed to the instrument 608, end-effector 112, or other object to be tracked, for example, with an array 612. Usually three or four markers 118, 804 are used with an array 612. The array 612 may include a linear section, a cross piece, and may be asymmetric such that the markers 118, 804 are at different relative positions and locations with respect to one another. For example, as shown in FIG. 13C, a probe 608A with a 4-marker tracking array 612 is shown, and FIG. 13B depicts the end-effector 112 with a different 4-marker tracking array 612.

In FIG. 13C, the tracking array 612 functions as the handle 620 of the probe 608A. Thus, the four markers 804 are attached to the handle 620 of the probe 608A, which is out of the way of the shaft 622 and tip 624. Stereophotogrammetric tracking of these four markers 804 allows the instrument 608 to be tracked as a rigid body and for the tracking system 100, 300, 600 to precisely determine the position of the tip 624 and the orientation of the shaft 622 while the probe 608A is moved around in front of tracking cameras 200, 326.

To enable automatic tracking of one or more tools 608, end-effector 112, or other object to be tracked in 3D (e.g., multiple rigid bodies), the markers 118, 804 on each tool 608, end-effector 112, or the like, are arranged asymmetrically with a known inter-marker spacing. The reason for asymmetric alignment is so that it is unambiguous which marker 118, 804 corresponds to a particular location on the rigid body and whether markers 118, 804 are being viewed from the front or back, i.e., mirrored. For example, if the markers 118, 804 were arranged in a square on the tool 608 or end-effector 112, it would be unclear to the system 100, 300, 600 which marker 118, 804 corresponded to which corner of the square. For example, for the probe 608A, it would be unclear which marker 804 was closest to the shaft 622. Thus, it would be unknown which way the shaft 622 was extending from the array 612. Accordingly, each array 612 and thus each tool 608, end-effector 112, or other object to be tracked should have a unique marker pattern to allow it to be distinguished from other tools 608 or other objects being tracked. Asymmetry and unique marker patterns allow the system 100, 300, 600 to detect individual markers 118, 804 then to check the marker spacing against a stored template to determine which tool 608, end effector 112, or other object they represent. Detected markers 118, 804 can then be sorted automatically and assigned to each tracked object in the correct order. Without this information, rigid body calculations could not then be performed to extract key geometric information, for example, such as tool tip 624 and alignment of the shaft 622, unless the user manually specified which detected marker 118, 804 corresponded to which position on each rigid body. These concepts are commonly known to those skilled in the methods of 3D optical tracking.

Turning now to FIGS. 14A-14D, an alternative version of an end-effector 912 with moveable tracking markers 918A-918D is shown. In FIG. 14A, an array with moveable tracking markers 918A-918D are shown in a first configuration, and in FIG. 14B the moveable tracking markers 918A-918D are shown in a second configuration, which is angled relative to the first configuration. FIG. 14C shows the template of the tracking markers 918A-918D, for example, as seen by the cameras 200, 326 in the first configuration of FIG. 14A; and FIG. 14D shows the template of tracking markers 918A-918D, for example, as seen by the cameras 200, 326 in the second configuration of FIG. 14B.

In this embodiment, 4-marker array tracking is contemplated wherein the markers 918A-918D are not all in fixed position relative to the rigid body and instead, one or more of the array markers 918A-918D can be adjusted, for example, during testing, to give updated information about the rigid body that is being tracked without disrupting the process for automatic detection and sorting of the tracked markers 918A-918D.

When tracking any tool, such as a guide tube 914 connected to the end effector 912 of a robot system 100, 300, 600, the tracking array's primary purpose is to update the position of the end effector 912 in the camera coordinate system. When using the rigid system, for example, as shown in FIG. 13B, the array 612 of reflective markers 118 rigidly extend from the guide tube 114. Because the tracking markers 118 are rigidly connected, knowledge of the marker locations in the camera coordinate system also provides exact location of the centerline, tip, and tail of the guide tube 114 in the camera coordinate system. Typically, information about the position of the end effector 112 from such an array 612 and information about the location of a target trajectory from another tracked source are used to calculate the required moves that must be input for each axis of the robot 102 that will move the guide tube 114 into alignment with the trajectory and move the tip to a particular location along the trajectory vector.

Sometimes, the desired trajectory is in an awkward or unreachable location, but if the guide tube 114 could be swiveled, it could be reached. For example, a very steep trajectory pointing away from the base 106 of the robot 102 might be reachable if the guide tube 114 could be swiveled upward beyond the limit of the pitch (wrist up-down angle) axis, but might not be reachable if the guide tube 114 is attached parallel to the plate connecting it to the end of the wrist. To reach such a trajectory, the base 106 of the robot 102 might be moved or a different end effector 112 with a different guide tube attachment might be exchanged with the working end effector. Both of these solutions may be time consuming and cumbersome.

As best seen in FIGS. 14A and 14B, if the array 908 is configured such that one or more of the markers 918A-918D are not in a fixed position and instead, one or more of the markers 918A-918D can be adjusted, swiveled, pivoted, or moved, the robot 102 can provide updated information about the object being tracked without disrupting the detection and tracking process. For example, one of the markers 918A-918D may be fixed in position and the other markers 918A-918D may be moveable; two of the markers 918A-918D may be fixed in position and the other markers 918A-918D may be moveable; three of the markers 918A-918D may be fixed in position and the other marker 918A-918D may be moveable; or all of the markers 918A-918D may be moveable.

In the embodiment shown in FIGS. 14A and 14B, markers 918A, 918 B are rigidly connected directly to a base 906 of the end-effector 912, and markers 918C, 918D are rigidly connected to the tube 914. Similar to array 612, array 908 may be provided to attach the markers 918A-918D to the end-effector 912, instrument 608, or other object to be tracked. In this case, however, the array 908 is comprised of a plurality of separate components. For example, markers 918A, 918B may be connected to the base 906 with a first array 908A, and markers 918C, 918D may be connected to the guide tube 914 with a second array 908B. Marker 918A may be affixed to a first end of the first array 908A and marker **918**B may be separated a linear distance and affixed to a second end of the first array **908**A. While first array **908** is substantially linear, second array **908**B has a bent or V-shaped configuration, with respective root ends, connected to the guide tube **914**, and diverging therefrom to distal ends in a V-shape with marker **918**C at one distal end and marker **918**D at the other distal end. Although specific configurations are exemplified herein, it will be appreciated that other asymmetric designs including different numbers and types of arrays **908**A, **908**B and different arrangements, numbers, and types of markers **918**A-**918**D are contemplated.

The guide tube **914** may be moveable, swivelable, or pivotable relative to the base **906**, for example, across a hinge **920** or other connector to the base **906**. Thus, markers **918**C, **918**D are moveable such that when the guide tube **914** pivots, swivels, or moves, markers **918**C, **918**D also pivot, swivel, or move. As best seen in FIG. **14**A, guide tube **914** has a longitudinal axis **916** which is aligned in a substantially normal or vertical orientation such that markers **918**A-**918**D have a first configuration. Turning now to FIG. **14**B, the guide tube **914** is pivoted, swiveled, or moved such that the longitudinal axis **916** is now angled relative to the vertical orientation such that markers **918**A-**918**D have a second configuration, different from the first configuration.

In contrast to the embodiment described for FIGS. **14**A-**14**D, if a swivel existed between the guide tube **914** and the arm **104** (e.g., the wrist attachment) with all four markers **918**A-**918**D remaining attached rigidly to the guide tube **914** and this swivel was adjusted by the user, the robotic system **100**, **300**, **600** would not be able to automatically detect that the guide tube **914** orientation had changed. The robotic system **100**, **300**, **600** would track the positions of the marker array **908** and would calculate incorrect robot axis moves assuming the guide tube **914** was attached to the wrist (the robot arm **104**) in the previous orientation. By keeping one or more markers **918**A-**918**D (e.g., two markers **918**C, **918**D) rigidly on the tube **914** and one or more markers **918**A-**918**D (e.g., two markers **918**A, **918**B) across the swivel, automatic detection of the new position becomes possible and correct robot moves are calculated based on the detection of a new tool or end-effector **112**, **912** on the end of the robot arm **104**.

One or more of the markers **918**A-**918**D are configured to be moved, pivoted, swiveled, or the like according to any suitable means. For example, the markers **918**A-**918**D may be moved by a hinge **920**, such as a clamp, spring, lever, slide, toggle, or the like, or any other suitable mechanism for moving the markers **918**A-**918**D individually or in combination, moving the arrays **908**A, **908**B individually or in combination, moving any portion of the end-effector **912** relative to another portion, or moving any portion of the tool **608** relative to another portion.

As shown in FIGS. **14**A and **14**B, the array **908** and guide tube **914** may become reconfigurable by simply loosening the clamp or hinge **920**, moving part of the array **908**A, **908**B relative to the other part **908**A, **908**B, and retightening the hinge **920** such that the guide tube **914** is oriented in a different position. For example, two markers **918**C, **918**D may be rigidly interconnected with the tube **914** and two markers **918**A, **918**B may be rigidly interconnected across the hinge **920** to the base **906** of the end-effector **912** that attaches to the robot arm **104**. The hinge **920** may be in the form of a clamp, such as a wing nut or the like, which can be loosened and retightened to allow the user to quickly switch between the first configuration (FIG. **14**A) and the second configuration (FIG. **14**B).

The cameras **200**, **326** detect the markers **918**A-**918**D, for example, in one of the templates identified in FIGS. **14**C and **14**D. If the array **908** is in the first configuration (FIG. **14**A) and tracking cameras **200**, **326** detect the markers **918**A-**918**D, then the tracked markers match Array Template **1** as shown in FIG. **14**C. If the array **908** is the second configuration (FIG. **14**B) and tracking cameras **200**, **326** detect the same markers **918**A-**918**D, then the tracked markers match Array Template **2** as shown in FIG. **14**D. Array Template **1** and Array Template **2** are recognized by the system **100**, **300**, **600** as two distinct tools, each with its own uniquely defined spatial relationship between guide tube **914**, markers **918**A-**918**D, and robot attachment. The user could therefore adjust the position of the end-effector **912** between the first and second configurations without notifying the system **100**, **300**, **600** of the change and the system **100**, **300**, **600** would appropriately adjust the movements of the robot **102** to stay on trajectory.

In this embodiment, there are two assembly positions in which the marker array matches unique templates that allow the system **100**, **300**, **600** to recognize the assembly as two different tools or two different end effectors. In any position of the swivel between or outside of these two positions (namely, Array Template **1** and Array Template **2** shown in FIGS. **14**C and **14**D, respectively), the markers **918**A-**918**D would not match any template and the system **100**, **300**, **600** would not detect any array present despite individual markers **918**A-**918**D being detected by cameras **200**, **326**, with the result being the same as if the markers **918**A-**918**D were temporarily blocked from view of the cameras **200**, **326**. It will be appreciated that other array templates may exist for other configurations, for example, identifying different instruments **608** or other end-effectors **112**, **912**, etc.

In the embodiment described, two discrete assembly positions are shown in FIGS. **14**A and **14**B. It will be appreciated, however, that there could be multiple discrete positions on a swivel joint, linear joint, combination of swivel and linear joints, pegboard, or other assembly where unique marker templates may be created by adjusting the position of one or more markers **918**A-**918**D of the array relative to the others, with each discrete position matching a particular template and defining a unique tool **608** or end-effector **112**, **912** with different known attributes. In addition, although exemplified for end effector **912**, it will be appreciated that moveable and fixed markers **918**A-**918**D may be used with any suitable instrument **608** or other object to be tracked.

When using an external 3D tracking system **100**, **300**, **600** to track a full rigid body array of three or more markers attached to a robot's end effector **112** (for example, as depicted in FIGS. **13**A and **13**B), it is possible to directly track or to calculate the 3D position of every section of the robot **102** in the coordinate system of the cameras **200**, **326**. The geometric orientations of joints relative to the tracker are known by design, and the linear or angular positions of joints are known from encoders for each motor of the robot **102**, fully defining the 3D positions of all of the moving parts from the end effector **112** to the base **116**. Similarly, if a tracker were mounted on the base **106** of the robot **102** (not shown), it is likewise possible to track or calculate the 3D position of every section of the robot **102** from base **106** to end effector **112** based on known joint geometry and joint positions from each motor's encoder.

In some situations, it may be desirable to track the positions of all segments of the robot **102** from fewer than three markers **118** rigidly attached to the end effector **112**. Specifically, if a tool **608** is introduced into the guide tube

114, it may be desirable to track full rigid body motion of the robot **902** with only one additional marker **118** being tracked.

Turning now to FIGS. **15A-15E**, an alternative version of an end-effector **1012** having only a single tracking marker **1018** is shown. End-effector **1012** may be similar to the other end-effectors described herein, and may include a guide tube **1014** extending along a longitudinal axis **1016**. A single tracking marker **1018**, similar to the other tracking markers described herein, may be rigidly affixed to the guide tube **1014**. This single marker **1018** can serve the purpose of adding missing degrees of freedom to allow full rigid body tracking and/or can serve the purpose of acting as a surveillance marker to ensure that assumptions about robot and camera positioning are valid.

The single tracking marker **1018** may be attached to the robotic end effector **1012** as a rigid extension to the end effector **1012** that protrudes in any convenient direction and does not obstruct the surgeon's view. The tracking marker **1018** may be affixed to the guide tube **1014** or any other suitable location of on the end-effector **1012**. When affixed to the guide tube **1014**, the tracking marker **1018** may be positioned at a location between first and second ends of the guide tube **1014**. For example, in FIG. **15**A, the single tracking marker **1018** is shown as a reflective sphere mounted on the end of a narrow shaft **1017** that extends forward from the guide tube **1014** and is positioned longitudinally above a mid-point of the guide tube **1014** and below the entry of the guide tube **1014**. This position allows the marker **1018** to be generally visible by cameras **200**, **326** but also would not obstruct vision of the surgeon **120** or collide with other tools or objects in the vicinity of surgery. In addition, the guide tube **1014** with the marker **1018** in this position is designed for the marker array on any tool **608** introduced into the guide tube **1014** to be visible at the same time as the single marker **1018** on the guide tube **1014** is visible.

As shown in FIG. **15**B, when a snugly fitting tool or instrument **608** is placed within the guide tube **1014**, the instrument **608** becomes mechanically constrained in 4 of 6 degrees of freedom. That is, the instrument **608** cannot be rotated in any direction except about the longitudinal axis **1016** of the guide tube **1014** and the instrument **608** cannot be translated in any direction except along the longitudinal axis **1016** of the guide tube **1014**. In other words, the instrument **608** can only be translated along and rotated about the centerline of the guide tube **1014**. If two more parameters are known, such as (1) an angle of rotation about the longitudinal axis **1016** of the guide tube **1014**; and (2) a position along the guide tube **1014**, then the position of the end effector **1012** in the camera coordinate system becomes fully defined.

Referring now to FIG. **15**C, the system **100**, **300**, **600** should be able to know when a tool **608** is actually positioned inside of the guide tube **1014** and is not instead outside of the guide tube **1014** and just somewhere in view of the cameras **200**, **326**. The tool **608** has a longitudinal axis or centerline **616** and an array **612** with a plurality of tracked markers **804**. The rigid body calculations may be used to determine where the centerline **616** of the tool **608** is located in the camera coordinate system based on the tracked position of the array **612** on the tool **608**.

The fixed normal (perpendicular) distance DF from the single marker **1018** to the centerline or longitudinal axis **1016** of the guide tube **1014** is fixed and is known geometrically, and the position of the single marker **1018** can be tracked. Therefore, when a detected distance DD from tool centerline **616** to single marker **1018** matches the known fixed distance DF from the guide tube centerline **1016** to the single marker **1018**, it can be determined that the tool **608** is either within the guide tube **1014** (centerlines **616**, **1016** of tool **608** and guide tube **1014** coincident) or happens to be at some point in the locus of possible positions where this distance DD matches the fixed distance DF. For example, in FIG. **15**C, the normal detected distance DD from tool centerline **616** to the single marker **1018** matches the fixed distance DF from guide tube centerline **1016** to the single marker **1018** in both frames of data (tracked marker coordinates) represented by the transparent tool **608** in two positions, and thus, additional considerations may be needed to determine when the tool **608** is located in the guide tube **1014**.

Turning now to FIG. **15**D, programmed logic can be used to look for frames of tracking data in which the detected distance Dp from tool centerline **616** to single marker **1018** remains fixed at the correct length despite the tool **608** moving in space by more than some minimum distance relative to the single sphere **1018** to satisfy the condition that the tool **608** is moving within the guide tube **1014**. For example, a first frame F**1** may be detected with the tool **608** in a first position and a second frame F**2** may be detected with the tool **608** in a second position (namely, moved linearly with respect to the first position). The markers **804** on the tool array **612** may move by more than a given amount (e.g., more than 5 mm total) from the first frame F**1** to the second frame F**2**. Even with this movement, the detected distance Dp from the tool centerline vector C' to the single marker **1018** is substantially identical in both the first frame F**1** and the second frame F**2**.

Logistically, the surgeon **120** or user could place the tool **608** within the guide tube **1014** and slightly rotate it or slide it down into the guide tube **1014** and the system **100**, **300**, **600** would be able to detect that the tool **608** is within the guide tube **1014** from tracking of the five markers (four markers **804** on tool **608** plus single marker **1018** on guide tube **1014**). Knowing that the tool **608** is within the guide tube **1014**, all 6 degrees of freedom may be calculated that define the position and orientation of the robotic end effector **1012** in space. Without the single marker **1018**, even if it is known with certainty that the tool **608** is within the guide tube **1014**, it is unknown where the guide tube **1014** is located along the tool's centerline vector C' and how the guide tube **1014** is rotated relative to the centerline vector C'.

With emphasis on FIG. **15**E, the presence of the single marker **1018** being tracked as well as the four markers **804** on the tool **608**, it is possible to construct the centerline vector C' of the guide tube **1014** and tool **608** and the normal vector through the single marker **1018** and through the centerline vector C'. This normal vector has an orientation that is in a known orientation relative to the forearm of the robot distal to the wrist (in this example, oriented parallel to that segment) and intersects the centerline vector C' at a specific fixed position. For convenience, three mutually orthogonal vectors k', j', i' can be constructed, as shown in FIG. **15**E, defining rigid body position and orientation of the guide tube **1014**. One of the three mutually orthogonal vectors k' is constructed from the centerline vector C', the second vector j' is constructed from the normal vector through the single marker **1018**, and the third vector i' is the vector cross product of the first and second vectors k', j'. The robot's joint positions relative to these vectors k', j', i' are known and fixed when all joints are at zero, and therefore rigid body calculations can be used to determine the location of any section of the robot relative to these vectors k', j', i' when the robot is at a home position. During robot movement, if the positions of the tool markers **804** (while the tool **608** is in the guide tube **1014**) and the position of the single marker **1018** are detected from the tracking system, and angles/linear positions of each joint are known from encoders, then position and orientation of any section of the robot can be determined.

In some embodiments, it may be useful to fix the orientation of the tool **608** relative to the guide tube **1014**. For example, the end effector guide tube **1014** may be oriented in a particular position about its axis **1016** to allow machining or implant positioning. Although the orientation of anything attached to the tool **608** inserted into the guide tube **1014** is known from the tracked markers **804** on the tool **608**, the rotational orientation of the guide tube **1014** itself in the camera coordinate system is unknown without the additional tracking marker **1018** (or multiple tracking markers in other embodiments) on the guide tube **1014**. This marker **1018** provides essentially a "clock position" from −180° to +180° based on the orientation of the marker **1018** relative to the centerline vector C'. Thus, the single marker **1018** can provide additional degrees of freedom to allow full rigid body tracking and/or can act as a surveillance marker to ensure that assumptions about the robot and camera positioning are valid.

FIG. **16** is a block diagram of a method **1100** for navigating and moving the end-effector **1012** (or any other end-effector described herein) of the robot **102** to a desired target trajectory. Another use of the single marker **1018** on the robotic end effector **1012** or guide tube **1014** is as part of the method **1100** enabling the automated safe movement of the robot **102** without a full tracking array attached to the robot **102**. This method **1100** functions when the tracking cameras **200**, **326** do not move relative to the robot **102** (i.e., they are in a fixed position), the tracking system's coordinate system and robot's coordinate system are co-registered, and the robot **102** is calibrated such that the position and orientation of the guide tube **1014** can be accurately determined in the robot's Cartesian coordinate system based only on the encoded positions of each robotic axis.

For this method **1100**, the coordinate systems of the tracker and the robot must be co-registered, meaning that the coordinate transformation from the tracking system's Cartesian coordinate system to the robot's Cartesian coordinate system is needed. For convenience, this coordinate transformation can be a 4×4 matrix of translations and rotations that is well known in the field of robotics. This transformation will be termed Tcr to refer to "transformation-camera to robot". Once this transformation is known, any new frame of tracking data, which is received as x,y,z coordinates in vector form for each tracked marker, can be multiplied by the 4×4 matrix and the resulting x,y,z coordinates will be in the robot's coordinate system. To obtain Tcr, a full tracking array on the robot is tracked while it is rigidly attached to the robot at a location that is known in the robot's coordinate system, then known rigid body methods are used to calculate the transformation of coordinates. It should be evident that any tool **608** inserted into the guide tube **1014** of the robot **102** can provide the same rigid body information as a rigidly attached array when the additional marker **1018** is also read. That is, the tool **608** need only be inserted to any position within the guide tube **1014** and at any rotation within the guide tube **1014**, not to a fixed position and orientation. Thus, it is possible to determine Tcr by inserting any tool **608** with a tracking array **612** into the guide tube **1014** and reading the tool's array **612** plus the single marker **1018** of the guide tube **1014** while at the same time determining from the encoders on each axis the current location of the guide tube **1014** in the robot's coordinate system.

Logic for navigating and moving the robot **102** to a target trajectory is provided in the method **1100** of FIG. **16**. Before entering the loop **1102**, it is assumed that the transformation Tcr was previously stored. Thus, before entering loop **1102**, in step **1104**, after the robot base **106** is secured, greater than or equal to one frame of tracking data of a tool inserted in the guide tube while the robot is static is stored; and in step **1106**, the transformation of robot guide tube position from camera coordinates to robot coordinates Tcr is calculated from this static data and previous calibration data. Tcr should remain valid as long as the cameras **200**, **326** do not move relative to the robot **102**. If the cameras **200**, **326** move relative to the robot **102**, and Tcr needs to be re-obtained, the system **100**, **300**, **600** can be made to prompt the user to insert a tool **608** into the guide tube **1014** and then automatically perform the necessary calculations.

In the flowchart of method **1100**, each frame of data collected consists of the tracked position of the DRB **1404** on the patient **210**, the tracked position of the single marker **1018** on the end effector **1014**, and a snapshot of the positions of each robotic axis. From the positions of the robot's axes, the location of the single marker **1018** on the end effector **1012** is calculated. This calculated position is compared to the actual position of the marker **1018** as recorded from the tracking system. If the values agree, it can be assured that the robot **102** is in a known location. The transformation Tcr is applied to the tracked position of the DRB **1404** so that the target for the robot **102** can be provided in terms of the robot's coordinate system. The robot **102** can then be commanded to move to reach the target.

After steps **1104**, **1106**, loop **1102** includes step **1108** receiving rigid body information for DRB **1404** from the tracking system; step **1110** transforming target tip and trajectory from image coordinates to tracking system coordinates; and step **1112** transforming target tip and trajectory from camera coordinates to robot coordinates (apply Tcr). Loop **1102** further includes step **1114** receiving a single stray marker position for robot from tracking system; and step **1116** transforming the single stray marker from tracking system coordinates to robot coordinates (apply stored Tcr). Loop **1102** also includes step **1118** determining current location of the single robot marker **1018** in the robot coordinate system from forward kinematics. The information from steps **1116** and **1118** is used to determine step **1120** whether the stray marker coordinates from transformed tracked position agree with the calculated coordinates being less than a given tolerance. If yes, proceed to step **1122**, calculate and apply robot move to target x, y, z and trajectory. If no, proceed to step **1124**, halt and require full array insertion into guide tube **1014** before proceeding; step **1126** after array is inserted, recalculate Tcr; and then proceed to repeat steps **1108**, **1114**, and **1118**.

This method **1100** has advantages over a method in which the continuous monitoring of the single marker **1018** to verify the location is omitted. Without the single marker **1018**, it would still be possible to determine the position of the end effector **1012** using Tcr and to send the end-effector **1012** to a target location but it would not be possible to verify that the robot **102** was actually in the expected location. For example, if the cameras **200**, **326** had been bumped and Tcr was no longer valid, the robot **102** would move to an erroneous location. For this reason, the single marker **1018** provides value with regard to safety.

For a given fixed position of the robot **102**, it is theoretically possible to move the tracking cameras **200**, **326** to a new location in which the single tracked marker **1018** remains unmoved since it is a single point, not an array. In such a case, the system **100**, **300**, **600** would not detect any error since there would be agreement in the calculated and tracked locations of the single marker **1018**. However, once the robot's axes caused the guide tube **1012** to move to a new location, the calculated and tracked positions would disagree and the safety check would be effective.

The term "surveillance marker" may be used, for example, in reference to a single marker that is in a fixed location relative to the DRB **1404**. In this instance, if the DRB **1404** is bumped or otherwise dislodged, the relative location of the surveillance marker changes and the surgeon **120** can be alerted that there may be a problem with navigation. Similarly, in the embodiments described herein, with a single marker **1018** on the robot's guide tube **1014**, the system **100**, **300**, **600** can continuously check whether the cameras **200**, **326** have moved relative to the robot **102**. If registration of the tracking system's coordinate system to the robot's coordinate system is lost, such as by cameras **200**, **326** being bumped or malfunctioning or by the robot malfunctioning, the system **100**, **300**, **600** can alert the user and corrections can be made. Thus, this single marker **1018** can also be thought of as a surveillance marker for the robot **102**.

It should be clear that with a full array permanently mounted on the robot **102** (e.g., the plurality of tracking markers **702** on end-effector **602** shown in FIGS. **7**A-**7**C) such functionality of a single marker **1018** as a robot surveillance marker is not needed because it is not required that the cameras **200**, **326** be in a fixed position relative to the robot **102**, and Ter is updated at each frame based on the tracked position of the robot **102**. Reasons to use a single marker **1018** instead of a full array are that the full array is more bulky and obtrusive, thereby blocking the surgeon's view and access to the surgical field **208** more than a single marker **1018**, and line of sight to a full array is more easily blocked than line of sight to a single marker **1018**.

Turning now to FIGS. **17**A-**17**B and **18**A-**18**B, instruments **608**, such as implant holders **608**B, **608**C, are depicted which include both fixed and moveable tracking markers **804**, **806**. The implant holders **608**B, **608**C may have a handle **620** and an outer shaft **622** extending from the handle **620**. The shaft **622** may be positioned substantially perpendicular to the handle **620**, as shown, or in any other suitable orientation. An inner shaft **626** may extend through the outer shaft **622** with a knob **628** at one end. Implant **10**, **12** connects to the shaft **622**, at the other end, at tip **624** of the implant holder **608**B, **608**C using typical connection mechanisms known to those of skill in the art. The knob **628** may be rotated, for example, to expand or articulate the implant **10**, **12**. U.S. Pat. Nos. 8,709,086 and 8,491,659, which are incorporated by reference herein, describe expandable fusion devices and methods of installation.

When tracking the tool **608**, such as implant holder **608**B, **608**C, the tracking array **612** may contain a combination of fixed markers **804** and one or more moveable markers **806** which make up the array **612** or is otherwise attached to the implant holder **608**B, **608**C. The navigation array **612** may include at least one or more (e.g., at least two) fixed position markers **804**, which are positioned with a known location relative to the implant holder instrument **608**B, **608**C. These fixed markers **804** would not be able to move in any orientation relative to the instrument geometry and would be useful in defining where the instrument **608** is in space. In addition, at least one marker **806** is present which can be attached to the array **612** or the instrument itself which is capable of moving within a pre-determined boundary (e.g., sliding, rotating, etc.) relative to the fixed markers **804**. The system **100**, **300**, **600** (e.g., the software) correlates the position of the moveable marker **806** to a particular position, orientation, or other attribute of the implant **10** (such as height of an expandable interbody spacer shown in FIGS. **17**A-**17**B or angle of an articulating interbody spacer shown in FIGS. **18**A-**18**B). Thus, the system and/or the user can determine the height or angle of the implant **10**, **12** based on the location of the moveable marker **806**.

In the embodiment shown in FIGS. **17**A-**17**B, four fixed markers **804** are used to define the implant holder **608**B and a fifth moveable marker **806** is able to slide within a pre-determined path to provide feedback on the implant height (e.g., a contracted position or an expanded position). FIG. **17**A shows the expandable spacer **10** at its initial height, and FIG. **17**B shows the spacer **10** in the expanded state with the moveable marker **806** translated to a different position. In this case, the moveable marker **806** moves closer to the fixed markers **804** when the implant **10** is expanded, although it is contemplated that this movement may be reversed or otherwise different. The amount of linear translation of the marker **806** would correspond to the height of the implant **10**. Although only two positions are shown, it would be possible to have this as a continuous function whereby any given expansion height could be correlated to a specific position of the moveable marker **806**.

Turning now to FIGS. **18**A-**18**B, four fixed markers **804** are used to define the implant holder **608**C and a fifth, moveable marker **806** is configured to slide within a pre-determined path to provide feedback on the implant articulation angle. FIG. **18**A shows the articulating spacer **12** at its initial linear state, and FIG. **18**B shows the spacer **12** in an articulated state at some offset angle with the moveable marker **806** translated to a different position. The amount of linear translation of the marker **806** would correspond to the articulation angle of the implant **12**. Although only two positions are shown, it would be possible to have this as a continuous function whereby any given articulation angle could be correlated to a specific position of the moveable marker **806**.

In these embodiments, the moveable marker **806** slides continuously to provide feedback about an attribute of the implant **10**, **12** based on position. It is also contemplated that there may be discreet positions that the moveable marker **806** must be in which would also be able to provide further information about an implant attribute. In this case, each discreet configuration of all markers **804**, **806** correlates to a specific geometry of the implant holder **608**B, **608**C and the implant **10**, **12** in a specific orientation or at a specific height. In addition, any motion of the moveable marker **806** could be used for other variable attributes of any other type of navigated implant.

Although depicted and described with respect to linear movement of the moveable marker **806**, the moveable marker **806** should not be limited to just sliding as there may be applications where rotation of the marker **806** or other movements could be useful to provide information about the implant **10**, **12**. Any relative change in position between the set of fixed markers **804** and the moveable marker **806** could be relevant information for the implant **10**, **12** or other device. In addition, although expandable and articulating implants **10**, **12** are exemplified, the instrument **608** could work with other medical devices and materials, such as spacers, cages, plates, fasteners, nails, screws, rods, pins, wire structures, sutures, anchor clips, staples, stents, bone grafts, biologics, or the like.

One aspect of the present invention related to determining the 3-dimensional position of an imaging arm of an imaging device for taking optimal images of a vertebral body will now be explained with reference to FIGS. **19-27**.

Most conventional systems do not have navigation capabilities and rely on users to position the C-arm. A few systems may have some navigation functions that allow a user to return to the previously stored position. In other words, existing systems may have the capability to let the user know where the imaging system may have been in the past. By contrast, the present invention as described with FIGS. **19-27** proposes to let the user know where the imaging system will need to be in the future to take optimal images.

FIG. **23** is an example of an x-ray imaging device **2300** having an automatic positioning capability with respect to the 3D position and orientation of its C-arm **2316** according to one aspect of the present invention. The imaging device **2300** includes a detector panel assembly **2314** containing a sensor array (not shown) for receiving x-ray transmission from an x-ray source **2312**. The imaging device **2300** is more fully described in U.S. patent Ser. No. 10/448,910 assigned to the applicant of the present invention, which is incorporate herein by reference. The imaging device **2300** is capable of communicating with the surgical robot system **300** through the connector panel **320** by a physical I/O cable or wirelessly through well-known wireless transmission methods including WiFi, Bluetooth and the like.

Unlike the imaging system **2300**, which does not require a calibration ring, manually operated C-arms such as **1308** typically will have a calibration ring **2200** mounted to the detector panel assembly **1316** as shown in FIG. **12**A. The calibration ring **2200** includes two spaced apart rings **2210**, **2212**, each with a planar surface. Each planar surface contains a plurality of radiopaque markers **2206**,**2208** that are spaced apart from each other in a selected pattern. Two sets of a plurality of circumferentially spaced optical markers **2202**,**2204** are also mounted to the rings **2210**,**2212**. The radiopaque markers **2206**,**2208** are used to perform an initial registration of the imaging device **1308** (i.e., mapping of the C-arm position and orientation relative to the patient from the imaging space to the camera coordinate system) so that the tracking subsystem **532** can track the position and orientation of the C-arm **1308** during the surgical procedure. For au automatically navigated imaging system **2300**, a calibration ring is unnecessary and tracking and navigation of the system can be done with optical markers **2310** or the encoders that are positioned in every moving part of the system. The encoders can be used to mark the relative location and orientation of the C-arm **2316** at any time in use.

Once initial registration has been performed, the cameras **326** of the tracking subsystem **532** can continuously track the C-arm **2316** position and orientation through the optical markers **2202**, **2204**, and optionally through the markers **2310** on the C-arm **2316** during the surgical procedure.

FIG. **21** illustrates a flowchart of a method of determining the 3-dimensional (3D) position and orientation of an imaging device for each vertebral level for taking optimal AP and lateral images so that only one set of images are needed. The processing steps in FIG. **21** can be performed by an image control module **409** in the computer **408**, processor of the imaging device **2300** itself, other remotely located processors or a combination thereof. In one embodiment, the image control module **409** includes computer executable code stored in a memory **410**.

In step **2100**, a user (typically an x-ray technician in the operating room) positions the imaging device **2300** around a patient table (not shown) such that the patient lying on the table is position inside the C-arm **2316**. Once the imaging device **2300** is positioned, a pair of x-ray images (one AP image and one lateral image) are taken by the user without regard to how accurately or optimally the C-arm **2316** is positioned so long as the vertebral levels of interest are included. A typical AP image **2602** and lateral image **2604** are shown in FIG. **26**. Once the two images are taken, they are received and stored by the computer **408**.

Along with the images, the computer **408** also receives and stores the 3D position and orientation of the C-arm (e.g., 3D position and orientation of the imaging panel/intensifier or x-ray source of the C-arm, or both) for each of the two images **2602**,**2604**.

In step **2102**, the vertebral bodies of interest are segmented for later analysis. Segmentation is a process by which certain points or features on a body part such as a vertebral body are identified. It can be manual, semi-automatic or fully automatic. An illustration of segmented vertebral bodies is shown in FIG. **26**. The semi-automatic or fully automatic segmentation methods for identifying the relevant points of the vertebral body are well-known in the art. For example, an open-source software program called "ITK-SNAP" (available at www.itksnap.net) may allow the user to interactively segment out each vertebral body.

Step **2102** may also identify the vertebral levels as part of the segmentation process. This identification process can be totally manual, which requires a user to identify each level. Alternatively, the identification process can also be semi-automatic or fully automatic. In a semi-automatic case, the user may identify at least one level and the remaining levels are automatically applied based on image processing. For example, once the user identifies one vertebral body as being L4 (as shown in FIG. **26**), the computer **408** automatically identifies all other levels based on the lordotic angle of the segmented bodies, for example. As a double check, the computer **408** may ask the user to confirm that the automatically identified levels, either by semi-automatic or fully automatic process, are correct.

In step **2104**, the computer **408** asks the user to identify which vertebral levels are of interest. The user then identifies them using a graphical user interface, for example by touching the displayed levels on a touch screen display device **304** (e.g., four levels from L1 to L4).

In step **2106**, the computer **408** retrieves from a database a 3D model **2702** of the spine including the vertebral bodies of interest. The 3D model may be based on a statistical model which is not specific to any patient as most spines generally follow a standard pattern or it could be based on a specific patient in question from a 3D scan. Alternatively, the standard 3D model can be enhanced by patient specific data such as the lordotic and kyphotic angles which are derived from the images **2602**,**2604**. The retrieved vertebral bodies **2702** are then scaled so that the size of the bodies are the same as those in the AP and lateral images. The scaling may be based on the segmentation information obtained from step **2102**.

In step **2106**, for each vertebral body of interest, the computer **408** performs an alignment of the retrieved 3D model of a selected vertebral body to the corresponding segmented vertebral body in the AP and lateral images **2602**,**2604**. One method that may be used is a "fluoro-CT merge", for example. One algorithm for the fluoro-CT merge can be found in an article entitled "Image-Assisted Navigation System for Spinal Surgery", Applied Bionics and Biomechanics, Volume 2015, Article ID 478062, 9 pages, published May 28, 2015 (downloaded from http://dx-.doi.org/10.1155/2015/478062), which is incorporated herein by reference. Essentially, the 3D vertebral model's position and orientation (including X, Y, Z, Yaw, Roll and Pitch) is adjusted by the computer **408** until an optimum alignment is achieved.

FIGS. **27**A and **27**B graphically illustrate the alignment method. FIG. **27**B illustrates the lateral test image **2710** from an x-ray source **2708**. The lateral test image **2710** and x-ray source **2706** respectively correspond to the detector panel **2710** in the detector panel assembly **2314** and x-ray source **2312** of the imaging device **2300** at the time the test image was taken. The AP test image **2710** and x-ray source **2706** respectively correspond to the detector panel **2710** and x-ray source **2312** of the imaging device **2300** at the time the test image was taken. As can be seen, the model vertebral body **2702** is scaled and manipulated until the body matches most closely aligns with the corresponding vertebral body in the AP and lateral images.

Step **2106** is repeated for each vertebral body of interest as identified in step **2104**.

Then, in step **2110**, based on the optimal 3D position and orientation of the vertebral body as determined in step **2108**, the computer **408** determines the optimal C-arm **2316** orientation and position (e.g., 3D position and orientation of either the detector panel **2314** or the x-ray source **2312**, or both) so as to center the vertebral body with perfect AP and lateral angles. Then, the determined optimal C-arm **2316** orientation and position for the vertebral body are stored in the memory **410**.

This optimal C-arm **2316** orientation and position determination of step **2110** can be partially seen in FIG. **27**A. FIG. **27**A shows the model vertebral body **2702** which has been aligned with the corresponding vertebral body in the AP and lateral test images **2602**,**2604**. As can be readily seen from the left screen shot **2720** showing a test AP image **2602**, an optimum position for the C-arm **2316** would include rotating it clockwise by about 15 degrees and moving it down by about half a vertebral level to center the vertebral body. From the right screen shot **2722** showing a lateral test image **2604**, an optimum position for the C-arm **2316** would include rotating it counter-clockwise by about 10 degrees and to move it left by about half a vertebral level to center the vertebral body in the image.

The optimal C-arm **2316** orientation and position of the vertebral body are then stored in the memory. In one embodiment, the orientation and position information for taking one of the two images are stored. Then, taking the other image is just a matter of rotating the C-arm **2316** by 90 degrees. In an alternative embodiment, the orientation and position information for taking both AP and lateral images are stored. If there are any additional levels that have not been processed, then steps **2106**-**2110** may be repeated.

In step **2112**, the computer **408** displays the available vertebral levels for optimal imaging for user selection in the display device **304**, one example of which is illustrated in FIG. **24**A. For each level, the display **304** displays two user input buttons **2402** and **2404**. These buttons are used to position the imaging system **2300** to the ideal or optimal imaging position. The positioning can be either manual or automatic, depending on the image equipment being used. Button **2402** is for taking an AP image and box **2404** is for taking a lateral image. The user can select the image to be taken by an input device such as a mouse, touchscreen or keyboard. In one embodiment, the user can make the selection by touching the input button through a touch-sensitive screen of the display device **304**.

In decision **2114**, the computer **408** determine whether the imaging device **2300** has an automatic positioning capability. The automatic positioning capability allows the computer **408** to send position and orientation commands to move and rotate the C-arm **2316** of the imaging device **2300** in an optimal 3D position and orientation as determined in step **2110**.

If the imaging device **2300** is determined to have such a capability, then control passes to step **2116**. In step **2116**, the computer **408** sends the optimal 3D position and orientation of the C-arm **2316** to the imaging device.

In one embodiment, the computer **408** sends an absolute position and orientation data to the imaging device **2300**. This is possible if the imaging device **2300** knows its exact position within the operating room. In another embodiment, the computer **408** sends movement instructions that incrementally moves and positions the C-arm **2316** step by step. The computer **408** knows the relative position of the C-arm **2316** from the initial registration of the imaging device **2300** to the patient. From the registration data and the optical markers **2310** on the gantry, the computer **408** can track the relative location and orientation of the C-arm **2316** relative to the patient. From the tracking data and while the markers are being tracked, the computer can issue a series of incremental positioning commands to the imaging device **2300** until optimal C-arm **2316** position and orientation are reached.

If the imaging device is determined not to have such an automatic positioning capability (such as imaging system **1304** of FIG. **12**A) in step **2114**, then control passes to step **2118**. In step **2118**, the computer **408** graphically displays on the display device **304** an indication of the C-arm **1308** position relative to the optimal position, and lets the user move and orient the C-arm **1308**. As the C-arm **1308** is moved by the user, the graphical display on the display device **304** is continuously updated to show the user how close the C-arm **1308** is to its optimal position. The location of the C-arm **1308** can be tracked by the optical markers **2202**,**2204** on the calibration ring **2200** or some other trackable markers that are positioned on the C-arm. One example of the graphical display is illustrated in FIG. **25**.

The left image displays an x-y-z coordinate of the C-arm **1308**. The dotted circle represents the optimal position of the C-arm **1308**. The center of the dotted circle represents the optimal X-Y position with the size of the dotted circle representing the optimal Z position. The solid circle represents the actual position of the C-arm **1308**. As the user moves the C-arm **1308**, the solid circle moves and changes its size to indicate its actual 3D (X-Y-Z) position relative to the optimal position.

The right image displays a Yaw-Pitch-Roll coordinate of the C-arm **1308**. The dotted circle represents the optimal orientation of the C-arm **1308**. The center of the dotted circle represents the optimal Yaw-Pitch position with the size of the dotted circle representing the optimal Roll position. The solid circle represents the actual position of the C-arm **1308** in terms of Yaw, Pitch and Roll. As the user moves the C-arm **1308**, the solid circle moves and changes its size to indicate its 3D orientation relative to the optimal position.

Once the solid circle on both coordinates have been aligned with the respective dotted circles, the imaging device **1304** is ready to take the appropriate image. For example, if L1-AP **2402** had been selected by the user, the imaging device **2300** takes the AP image. It can be done by actuating an appropriate button on the imaging device or instructions from the computer **408** can be sent to do so.

Alternatively, once a vertebral level is selected, the computer **408** can send instructions to the imaging device **2300** to take both the optimal AP and lateral images based on the stored optimal position and orientation that have been determined through steps **2102-2110**.

In one embodiment, for every image taken and stored by the imaging device **2300**, the computer **408** also stores in the memory **410** the image as well as the position and orientation information of the C-arm **2316**. That can be achieved either through the optical markers **2310** and **2202-2204**, or through the imaging device's internal positioning elements such as encoders in the motors controlling every axis and 3D position of the C-arm **2316**.

As the images are being taken, additional vertebral levels may become available in the newly acquired images. For example, as optimal L1 images are being taken, those images may contain new levels such as L4. In one aspect of the present invention, the computer **408** stores the newly acquired images (both AP and lateral) and their position and orientation in the memory **410** and then repeats steps **2102** through **2112** for the new vertebral level if the new level was identified in step **2104** as of interest.

In another aspect of the present invention, the computer **408** may refine the optimal 3D position and orientation data which have already been obtained. In the same example, the computer **408** may repeat steps **2102** through **2112** for L2 and L3 based on the newly acquired AP and lateral images. Since the images were taken based on the optimal 3D position and orientation data for L1, they may also contain a more optimally aligned levels for L2 and L3. Thus, the refined 3D and orientation positions for L2 and L3 will likely be even more accurate than before.

In step **2112**, in addition to input buttons **2402**,**2404** for the old levels (e.g., L1-L3), the computer **408** displays the graphical representation of input buttons **2410**,**2412** for the new level (e.g., AP and lateral for L4) on the display device **304**.

As can be appreciated, the method described above substantially reduces the setup time for positioning an x-ray imaging device in the operating room as only two fluoro shots (one set of AP and lateral images) are needed for each vertebral level, instead of requiring 10 or more. This advantageous feature yields many benefits including a substantial reduction in procedure time, substantial reduction in radiation exposure for the patient as well as the medical professionals, and reduced cost for the procedure due to less time being required for the procedures. Perhaps more importantly, because the present invention allows more optimal images to be taken, it allows the physician to place the implants more accurately, which leads to better patient outcome in many surgeries.

Although several embodiments of the invention have been disclosed in the foregoing specification, it is understood that many modifications and other embodiments of the invention will come to mind to which the invention pertains, having the benefit of the teaching presented in the foregoing description and associated drawings. It is thus understood that the invention is not limited to the specific embodiments disclosed hereinabove, and that many modifications and other embodiments are intended to be included within the scope of the appended claims. For example, while the invention has been described with reference to lumbar spine, it can applicable to any body implant (with virtyual 3-D model of the implant), any body structure or other other areas of the spine including cervical spine and thoracic spine as well as any other body part that requires A-P and lateral images such as knees It is further envisioned that features from one embodiment may be combined or used with the features from a different embodiment described herein. Moreover, although specific terms are employed herein, as well as in the claims which follow, they are used only in a generic and descriptive sense, and not for the purposes of limiting the described invention, nor the claims which follow. The entire disclosure of each patent and publication cited herein is incorporated by reference, as if each such patent or publication were individually incorporated by reference herein. Various features and advantages of the invention are set forth in the following claims.

What is claimed is:

1. A method of determining the 3-dimensional position and orientation of an imaging arm of an x-ray imaging device for taking optimal x-ray images of a vertebral body, the method comprising:

receiving, from the x-ray imaging device, test images including:

a first x-ray image of the vertebral body; and

a second x-ray image of the vertebral body at a different angle than the first x-ray image;

segmenting, among a plurality of vertebral bodies contained in the x-ray test images, vertebral bodies such that known reference points are identified;

identifying a vertebral body among the plurality of vertebral bodies contained in the x-ray test images such that a level of the vertebral body is identified;

retrieving a 3-dimensional model of the vertebral body from a storage device;

aligning the retrieved 3-dimensional model against the identified vertebral body contained in the test images by performing a fluoro-3D merge;

automatically determining a 3-dimensional position and orientation of the imaging arm of the x-ray device for taking future optimal A-P and lateral x-ray images based on the aligned 3-dimensional model.

2. The method of claim **1**, wherein:

the step of identifying includes identifying a plurality of vertebral bodies contained in the test images;

for each identified vertebral body, repeating the steps of 3-dimensionally aligning and determining the 3-dimensional position and orientation of the imaging arm.

3. The method of claim **1**, wherein:

the step of identifying includes segmenting the vertebral body in the test images; and

the step of 3-dimensionally aligning includes 3-dimensionally aligning the retrieved 3-dimensional model against the segmented vertebral body.

4. The method of claim **1**, wherein the step of aligning includes scaling the size of the 3-dimensional model of the vertebral body to correspond to the identified vertebral body contained in the test images.

5. The method of claim **1**, wherein the step of 3-dimensionally aligning includes performing a fluoro-CT merge.

6. The method of claim **1**, further comprising:

receiving the optimal A-P and lateral x-ray images based on the determined 3-dimensional position and orientation of the imaging arm;

identifying an additional vertebral body contained in the received optimal A-P and lateral x-ray images;

retrieving a 3-dimensional model of the identified additional vertebral body from the storage device;

3-dimensionally aligning the retrieved 3-dimensional model against the received optimal A-P and lateral x-ray images;

determining the 3-dimensional position and orientation of the imaging arm for taking optimal A-P and lateral x-ray images for the identified additional vertebral body based on the aligned 3-dimensional model of the additional vertebral body.

7. The method of claim **1**, prior to receiving test images, further comprising deriving the 3-dimensional model from a 3-dimensional scan of the patient.

8. The method of claim **1**, further comprising:

tracking the position and orientation of the imaging arm through navigation markers; and

displaying in a display device a graphical representation of the extent of the imaging arm alignment relative to the optimal 3-dimensional position and orientation.

9. The method of claim **8**, wherein the step of displaying includes:

displaying a graphical representation of an optimal position and orientation of the imaging arm on a coordinate system;

displaying a graphical representation of a current position and orientation of the imaging arm on the same coordinate system;

continuously updating the location of the graphical representation of the current position of the imaging arm as the imaging arm is being moved.

10. The method of claim **9**, wherein the step of continuously displaying includes:

displaying a circle in the center of a coordinate system as the graphical representation of the optimal position of the imaging arm;

varying the size of the graphical representation of the current position of the imaging arm as the imaging arm moves in either X, Y or Z direction.

11. The method of claim **8**, wherein the step of displaying includes displaying graphical representation of how closely the imaging arm is aligned with the optimal 3-dimensional position and orientation of the imaging arm relative to the identified vertebral body to be imaged.

12. The method of claim **1**, further comprising:

continuously tracking the position of the imaging arm through the navigation markers; and

automatically sending instructions to the imaging device based on the determined 3-dimensional position and orientation of the imaging arm,

wherein the step of registering the image device to the tracking subsystem is performed without the use of any radio-opaque markers.

13. A method of determining the 3-dimensional position and orientation of an imaging arm of an x-ray imaging device for taking optimal x-ray images of a body part, the method comprising:

registering the imaging device to a tracking subsystem having one or more cameras configured to detect navigation markers located on the imaging device, wherein the navigation markers are optical markers,

receiving, from the x-ray imaging device, test images including:

a first x-ray image of the body part; and

a second x-ray image of the body part at a different angle than the first x-ray image;

segmenting, among a plurality of vertebral bodies contained in the test images, vertebral bodies such that known reference points are identified;

identifying a vertebral body among the plurality of vertebral bodies contained in the x-ray test images such that a level of the vertebral body is identified;

retrieving a generic non patient-specific 3-dimensional model of the body part from a storage device;

aligning the retrieved 3-dimensional model against the identified vertebral body contained in the test images by performing a fluoro-3D merge;

automatically determining a 3-dimensional position and orientation of the imaging arm of the imaging device for taking future optimal A-P and lateral x-ray images based on the aligned 3-dimensional model; and

taking the future optimal A-P and lateral x-ray images after the imaging arm has been adjusted to the determined 3-dimensional position and orientation.

14. The method of claim **13**, further comprising segmenting the body part in the test images, wherein the step of 3-dimensionally aligning includes 3-dimensionally aligning the retrieved 3-dimensional model against the segmented body part.

15. The method of claim **13**, wherein the step of aligning includes scaling the size of the 3-dimensional model of the body part to correspond to the body part contained in the test images.

16. The method of claim **13**, wherein the step of 3-dimensionally aligning includes performing a fluoro-CT merge.

17. The method of claim **13**, prior to receiving test images, further comprising deriving the 3-dimensional model from a 3-dimensional scan of the patient.

18. The method of claim **13**, further comprising:

tracking the position and orientation of the imaging arm through navigation markers; and

displaying in a display device a graphical representation of the extent of the imaging arm alignment relative to the optimal 3-dimensional position and orientation.

19. The method of claim **18**, wherein the step of displaying includes:

displaying a graphical representation of an optimal position and orientation of the imaging arm on a coordinate system;

displaying a graphical representation of a current position and orientation of the imaging arm on the same coordinate system;

continuously updating the location of the graphical representation of the current position of the imaging arm as the imaging arm is being moved.

20. The method of claim **19**, wherein the step of continuously displaying includes:

displaying a circle in the center of a coordinate system as the graphical representation of the optimal position of the imaging arm;

varying the size of the graphical representation of the current position of the imaging arm as the imaging arm moves in either X, Y or Z direction.

21. The method of claim **18**, wherein the step of displaying includes displaying graphical representation of how closely the imaging arm is aligned with the optimal 3-dimensional position and orientation of the imaging arm relative to the body part to be imaged.

22. The method of claim **13**, further comprising:

continuously tracking the position of the imaging arm through the navigation markers; and

automatically sending instructions to the imaging device based on the determined 3-dimensional position and orientation of the imaging arm.

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